

Brand Risk Analysis

PHASE I REPORT

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

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December 2026

BONAFIDE CERTIFICATE

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ABSTRACT

E-commerce recommendation systems play an important role in helping users discover relevant products by considering factors such as product relevance, popularity, and purchase intent. However, conventional recommendation approaches primarily focus on user-product interactions and do not sufficiently consider whether the behavioral patterns associated with product brands are consistent with those observed across the marketplace. This creates a gap in incorporating behavioral trust into recommendation ranking, particularly when unusual brand-level patterns may influence the reliability of recommendations. This project aims to develop a trust-aware recommendation system that integrates session-level purchase intent with brand-level behavioral suspiciousness, while clearly distinguishing anomalous behavior from confirmed fraud or legal wrongdoing.

The proposed system combines a User Intent Prediction Model with a Trust Aware Behavioral Suspiciousness Model to improve the ranking of recommended products. Session-level behavioral, product, category, price, and temporal features are used to estimate purchase probability through a stacking ensemble of machine-learning models, while validated brand-level behavioral features are analysed using an Isolation Forest to identify unusual patterns. Candidate products are generated through multiple recommendation channels and re-ranked using product relevance, purchase intent, and brand trust information with a soft risk-aware adjustment rather than automatic product removal. The approach is evaluated using standard top- K recommendation metrics and trust-related measures to assess both recommendation effectiveness and the influence of behavioral trust on the final ranking.

ACKNOWLEDGEMENT

Initially, we thank the Almighty for being with us through every walk of our lives and showering His blessings throughout the endeavour to complete this project report. We express our sincere thanks to our Chairman, **Mr. S. Meganathan, B.E., F.I.E.**, our Vice Chairman, **Mr. Abhay Shankar Meganathan, B.E., M.S.**, and our respected Chairperson, **Dr. Thangam Meganathan, Ph.D.**, for providing us with the requisite infrastructure and for their sincere endeavour in educating us in their premier institution. We express our sincere gratitude to our Principal, **Dr. S. N. Murugesan, M.E., Ph.D.**, for his kind support, encouragement, and provision of the necessary facilities to complete our project work successfully.

We convey our sincere and deepest gratitude to our Head of the Department, **Dr. J. Manoranjini, M.E., Ph.D.**, Professor and Head, Department of Computer Science and Engineering, for her valuable support, guidance, and encouragement throughout the project work. We express our sincere thanks to our Project Guide, **Ms. M. Shafiya Banu, M.E.**, Assistant Professor, Department of Computer Science and Engineering, for her valuable guidance, constant encouragement, constructive suggestions, and support throughout the course of this project. We extend our sincere gratitude to our Project Coordinator, **Dr.S.Madhusudhanan, M.E., Ph.D.**, Associate Professor, Department of Computer Science and Engineering, for his valuable coordination, guidance, and support during the development and completion of our project. Finally, we would like to thank all the faculty members and staff of the Department of Computer Science and Engineering, Rajalakshmi Engineering College, for their valuable support and encouragement throughout the project.

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CHAPTER 1

INTRODUCTION

1.1 Background

Electronic commerce platforms mediate a large number of short, goal-directed interactions. A customer may arrive through a search result, inspect several products, compare prices, move between categories, place an item in a cart, remove it, or leave without purchasing. Each action contributes to a clickstream: a time-ordered record of how the customer navigates and responds during the visit. Although any individual event provides limited information, the sequence and composition of events within a session can reveal the customer's current level of engagement and the direction of the shopping task. Clickstream analysis is therefore especially useful when a detailed long-term user profile is unavailable, incomplete, or no longer representative of the customer's immediate interests.

Recommendation systems reduce the effort required to search a large catalog by selecting a comparatively small set of products for presentation. Classical e-commerce applications have used popularity, association, content similarity, and customer history to support product selection and cross-selling [6]. Contemporary session-based methods give greater emphasis to the current visit because short-term interactions often contain the most direct evidence of what a customer presently wants. Research has shown that current-session interactions, contextual attributes, and item relationships can provide useful recommendation signals without depending entirely on a persistent identity [7, 8, 14].

Purchase-intention prediction complements recommendation but solves a different task. A recommender identifies which products should be presented, whereas an intention predictor estimates whether the observed session behaviour is consistent with a purchase outcome. Studies of online retail sessions have demonstrated the value of aggregate activity, recent actions, temporal characteristics, and learned session representations for early prediction [1, 2, 3, 4]. A continuous purchase probability can consequently express the strength of commercial intent in the current session and can be used as contextual information when ranking products.

Relevance and purchase intent, however, do not provide a complete account of a candidate product. A product can be highly relevant to the active session and can belong to a brand whose aggregated interaction pattern is unusual. Such behaviour may arise from many causes, including legitimate niche demand, small samples, promotional activity, catalog differences, or data-quality problems. It may also indicate behaviour that deserves further review. For that reason, unusual behaviour should be treated as suspiciousness evidence rather than as proof of fraud. The present project incorporates this evidence into ranking through a trust-related signal while preserving cautious interpretation.

1.2 E-Commerce Recommendation and the Trust Dimension

The practical objective of recommendation is not merely to predict the next event. It is to arrange available products in an order that supports customer discovery and platform objectives. In a conventional relevance-oriented pipeline, the products with the strongest co-occurrence, category match, or popularity

may rise to the highest positions. When a purchase probability is added, the current session’s intent can strengthen the priority of candidates that already appear relevant. These operations improve utility, but they remain largely trust-blind because they do not consider whether the brand associated with each product exhibits unusual aggregate behaviour.

Trust in this project has a deliberately limited operational meaning. It is not a subjective customer review score, a legal assessment, or a verified measure of seller integrity. It is a derived ranking signal based on brand-level behavioural consistency within the available clickstream evidence. The term is used because the signal controls how confidently the recommendation system promotes products associated with a brand. Stronger behavioural trust supports ranking priority, while higher suspiciousness reduces priority through a soft re-ranking mechanism. The product remains eligible, so relevance and purchase intent continue to contribute to the final order.

This interpretation is important for both technical correctness and responsible use. Unsupervised anomaly detection identifies observations that are easier to isolate from the rest of the data; it does not explain why those observations are unusual. Research on e-commerce fraud detection may use labels, graph structures, verified outcomes, or competitive neural models [11, 12]. The present work does not have equivalent verified fraud labels and does not claim to reproduce those systems. Instead, it uses anomaly analysis to construct behavioural suspiciousness that influences recommendation ranking without becoming a final accusation.

1.3 Clickstream Data and Shopping Sessions

The source data records event time, event type, product identifier, category information, brand, price, user identifier, and user-session identifier. The session identifier provides a natural unit for grouping actions that belong to the same visit. Ordering events by time within this group reconstructs the observable sequence through which the user explored the catalog. The resulting session can be summarized by counts, diversity measures, price statistics, duration, time-of-entry characteristics, and recent actions.

Session-level representation offers two advantages. First, it reduces a variable-length interaction sequence to a fixed set of features suitable for structured machine-learning models. Second, it separates the intent of the active visit from the long-term identity of the user. A customer may browse electronics during one session and household products during another; a session representation allows the recommendation process to react to that change. It also supports anonymous or new users for whom a historical profile is unavailable.

The purchase label requires careful construction. A session is labelled as a purchase session if it contains at least one purchase event. Once the label is known, purchase events are removed from the features supplied to the intent model. Retaining them would reveal the target directly and would cause data leakage. The model must learn from the behaviour preceding the outcome, not from the outcome itself. This design is consistent with the practical objective of estimating intent before the purchase has occurred.

1.4 Problem Statement

The central problem is the separation between recommendation utility and behavioural trust analysis. A relevance-based or intent-enhanced recommender can identify products that fit the current session but may still give high priority to products whose brands show unusual clickstream behaviour. Conversely, a stand-alone anomaly detector can identify unusual brand patterns but cannot influence what the customer sees unless its output is integrated into recommendation ranking. The project therefore addresses the following problem: how can current-session purchase intent and brand-level behavioural trust be incorporated into one recommendation pipeline while preserving product relevance, avoiding unsupported fraud conclusions, and retaining products through soft re-ranking rather than unconditional removal?

The problem includes several technical subproblems. Raw events must be cleaned and ordered into sessions. The session must be represented without purchase leakage. An imbalanced binary outcome must be predicted while retaining a calibrated continuous output for ranking. Brand activity must be summarized through validated behavioural features and standardized before unsupervised analysis. Candidate retrieval must provide sufficient coverage through complementary channels. Finally, model outputs with different meanings and granularities must be joined correctly: intent belongs to a session, while trust belongs to the brand of a candidate product.

1.5 Limitations of Existing Approaches

Existing approaches often optimize a single part of this chain. Popularity-based recommendation is simple and robust, but it can overemphasize frequently interacted products and offers limited personalization. Co-occurrence and session-based methods represent immediate interest more directly, yet their principal objective remains relevance. Purchase-intention models estimate conversion likelihood, but a high probability describes the session and says nothing about whether a particular brand exhibits consistent behaviour. Anomaly-detection systems can reveal unusual patterns, but their outputs are frequently used in a separate monitoring process rather than in recommendation.

Another limitation concerns the treatment of risk. A hard exclusion rule is easy to implement but can be disproportionate when behavioural evidence is uncertain. An unusual brand may be legitimate, newly introduced, highly specialized, affected by a promotion, or represented by incomplete data. Completely deleting all associated products can reduce catalog coverage and can turn an unsupervised signal into an unjustified decision. A graded response is more appropriate: the system can lower ranking priority according to behavioural risk while preserving the product in the candidate set.

Data sparsity and missing brand information introduce further difficulty. Some candidates are associated with unknown brands, and some brands have too little lifetime activity for reliable model-derived trust. Treating these cases as highly trusted would ignore uncertainty, while treating them as highly suspicious would overstate the available evidence. The implemented pipeline therefore uses neutral fallback values for integration when reliable brand-level information is absent. These fallbacks represent uncertainty and are not fraud labels.

1.6 Motivation

The motivation for the project is to develop a recommendation process that remains useful to the customer while making better use of behavioural evidence already present in clickstream data. Relevance, intent, and trust answer different questions. Relevance asks whether a product fits the session context. Purchase probability asks how strongly the session resembles one that will result in a transaction. Behavioural trust asks whether the product's brand demonstrates a comparatively consistent aggregate pattern. Combining these signals permits the ranking process to acknowledge both immediate preference and wider behavioural context.

The project is also motivated by interpretability. A final-year implementation should permit each major ranking influence to be traced to its source. The candidate relevance is produced by retrieval, purchase probability is produced by the User Intent Prediction Model, and trust information is produced by the TrustAware Behavioral Suspiciousness Model. This separation makes it possible to verify integration, inspect fallbacks, and explain why candidate positions changed without misrepresenting an anomaly score as a fraud probability.

1.7 Aim and Objectives

The aim is to design, implement, and evaluate an integrated trust-aware e-commerce recommendation system that uses clickstream-derived purchase intent and brand-level behavioural suspiciousness to re-rank relevant product candidates.

The objectives are:

1. to preprocess the e-commerce clickstream data, remove duplicates, handle missing brand information, order events chronologically, and form session-level records;
2. to construct purchase and non-purchase labels while removing purchase events from predictive input to prevent leakage;
3. to engineer aggregate, recent-action, price-related, and time-related session features;
4. to implement the User Intent Prediction Model using LightGBM, XGBoost, and Random Forest base learners with Logistic Regression as the stacking meta-classifier;
5. to retain continuous purchase probability for recommendation and use a threshold only for binary classification evaluation;
6. to implement the TrustAware Behavioral Suspiciousness Model using seven standardized brand-level features and Isolation Forest;
7. to interpret the brand output as behavioural suspiciousness and trust information rather than proof of fraud;
8. to retrieve candidates through co-view, related-category, and global-popularity channels and consolidate duplicates;
9. to integrate session intent and brand trust with candidate products and apply a soft trust-aware

re-ranking process;

10. to evaluate purchase prediction, trust coverage, recommendation relevance, ranking quality, average trust, and high-risk exposure; and
11. to document limitations and responsible future extensions without claiming unsupported deployment or detection capability.

1.8 Scope

The project uses the 2019-Oct.csv multi-category e-commerce behaviour dataset [15]. It focuses on clickstream events that support session-level purchase prediction, brand-level behavioural aggregation, product candidate retrieval, and offline recommendation evaluation. The validated catalog contains 30,223 products and 765 brands for which behavioural trust information is available. Recommendation evaluation uses 1,000 sessions and top-five and top-ten metrics.

The scope includes preprocessing, sessionization, feature engineering, model training, saved model output, candidate retrieval, model integration, ranking, backend access, and presentation through a demonstration interface. The evaluation is offline and does not establish production-scale latency, distributed persistence, causal effects on customer behaviour, or commercial uplift. Brand suspiciousness is based only on available behavioural evidence. Product reviews, returns, seller verification, delivery performance, and confirmed enforcement outcomes are outside the present dataset and are therefore outside the validated trust interpretation.

1.9 Proposed Solution

The proposed solution is one coherent pipeline with four connected stages. First, the User Intent Prediction Model processes session features and produces a continuous purchase probability. Second, the TrustAware Behavioral Suspiciousness Model processes standardized brand-level behavioural features and produces suspiciousness, trust information, and a behavioural-risk category. Third, candidate retrieval gathers products through co-view, related-category, and global-popularity evidence, removes duplicates, and assigns a bounded relevance value. Fourth, trust-aware re-ranking associates every candidate with the session probability and the trust record of its brand before ordering the list.

The ranking process is intentionally described conceptually. Product relevance remains the starting point, purchase probability expresses the strength of current-session intent, and brand trust adjusts the priority of the associated product. Low-risk brands receive little or no ranking reduction, while Medium- and High-risk behavioural categories receive progressively stronger soft reductions. The process modifies ordering rather than deleting products. This design retains useful candidates and avoids presenting an unsupervised anomaly result as a definitive exclusion decision.

1.10 Contributions

The first contribution is a leakage-controlled session representation that combines overall behaviour with the five most recent non-purchase actions. The second is a stacking-based continuous purchase-intent

estimator that combines three tree learners through a Logistic Regression meta-classifier. The third is a brand-level behavioural suspiciousness component that uses standardized features and Isolation Forest while explicitly separating anomalous behaviour from confirmed fraud. The fourth is the integration of these outputs with multi-channel candidate retrieval. The fifth is an evaluation that measures recommendation utility and behavioural composition together through HitRate, Precision, Recall, NDCG, Average Trust, and High-Risk Exposure.

1.11 Organization of the Report

Chapter 2 reviews research on e-commerce recommendation, session-aware methods, clickstream purchase prediction, tree-based learning, stacking, anomaly detection, and trust-related recommendation. Chapter 3 analyses the system problem and specifies functional and non-functional requirements. Chapter 4 presents the methodology, including preprocessing, session features, both implemented models, candidate retrieval, integration, and evaluation design. Chapter 5 explains the system design and implementation, using the supplied overall architecture as the only project figure. Chapter 6 reports the validated numerical results and discusses their meaning and limitations. Chapter 7 summarizes the contribution, identifies limitations, and proposes future work. The report concludes with the references supplied by the attached research paper.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter positions the project within research on recommendation and behavioural modelling. The review is organized by research problem rather than as a list of publications. It first considers e-commerce recommendation and session-aware retrieval, then examines clickstream purchase-intention prediction and structured machine learning. It subsequently discusses anomaly detection, behavioural suspiciousness, and trust-related recommendation. For each area, the review identifies what prior work contributes and what remains unresolved by that work in relation to an integrated trust-aware pipeline.

2.2 E-Commerce Recommendation Systems

Schafer, Konstan, and Riedl provided an influential account of recommendation applications in electronic commerce [6]. Their work described how recommendation supports product discovery, cross-selling, customer decision-making, and navigation through large catalogs. It also demonstrated that e-commerce recommendation is not a single algorithmic task: systems can use association, content, demographic information, ratings, or purchase histories depending on the available data and business setting. This broad view remains relevant because the present project also combines different signals rather than relying on one model output.

The limitation of a general e-commerce recommendation framework is that it does not by itself determine how short-term intent or behavioural trust should be represented. Popularity and historical association can be effective, but they may not respond quickly to a customer's current purpose. They also tend to treat product interaction volume as utility evidence without considering whether the brand-level pattern behind that volume is unusual. The present project builds on the product-discovery objective of e-commerce recommendation while adding session intent and behavioural trust as distinct ranking inputs.

Cabrera-Sánchez et al. examined factors influencing the use of online recommendation systems in e-commerce [13]. Their work emphasizes that recommendation success depends not only on predictive output but also on how users perceive and accept recommendations. This perspective supports attention to trust, though the meaning of trust in their study differs from the operational behavioural trust used here. The current system does not claim to measure customer psychology; instead, it creates a traceable model-derived signal that controls ranking priority. The connection is that recommendation quality is broader than raw relevance and can be influenced by confidence-related considerations.

2.3 Session-Based and Session-Aware Recommendation

Session-based recommendation addresses situations in which the current sequence of interactions is the primary source of preference information. Salampasis et al. proposed a flexible session-based recommender for e-commerce and considered co-occurrence-oriented and more advanced techniques [7]. Their work demonstrates the practical value of representing short-term activity and combining methods

suited to the characteristics of an e-commerce session. Co-occurrence is particularly relevant to the present candidate retrieval design because products viewed in related contexts can be retrieved without requiring a persistent customer profile.

Kersbergen, Sprangers, and Schelter studied low-latency session-based recommendation at e-commerce scale [8]. Their system-level focus is important because recommendation must often operate under serving constraints, not merely achieve offline accuracy. They show that production recommendation requires attention to data organization and efficient candidate delivery. The current project shares the objective of producing recommendations from session activity but validates an offline integrated pipeline and a local demonstration implementation. It does not claim the production scale or latency guarantees investigated in that work.

Bauer and Jannach examined hybrid session-aware recommendation using feature-based models [14]. Hybridization is relevant because a session signal can be strengthened by item and contextual features. Their findings motivate the use of multiple evidence sources, but the present project differs in the kind of additional information integrated. Rather than using behavioural suspiciousness as another relevance feature, the project preserves it as a separate brand-level trust signal. This separation improves conceptual clarity: session behaviour estimates customer intent, retrieval estimates product relevance, and brand analysis supplies behavioural trust.

Mo and Wang investigated popularity-stratified preference modelling for session-based recommendation [5]. Their work addresses the tendency of recommenders to overemphasize popular items and shows that popularity requires careful treatment within preference modelling. Global popularity serves a narrower purpose in the present system: it is one of three candidate-retrieval channels and improves coverage when session evidence is limited. It does not determine the final order alone. Co-view and related-category channels introduce session relevance, while intent and trust influence re-ranking.

Collectively, these studies confirm that current-session data can support recommendation and that multi-source designs can improve flexibility. Their principal limitation in relation to this project is that they do not directly connect a brand-level anomaly-derived trust signal with continuous purchase intent in the final candidate order. That integration constitutes the project-specific research and implementation gap.

2.4 Clickstream Analysis and Purchase-Intention Prediction

Clickstream purchase prediction seeks to infer an outcome from actions that occur before the transaction. Tokuc and Dag compared clickstream representations and machine-learning models for predicting user purchases [1]. Their analysis of aggregate, recent-action, and hybrid representations is directly relevant to the feature design used in this project. Aggregate features describe the overall session, while recent-action positions preserve limited sequential evidence. The combination avoids treating a session as a simple count vector and avoids the complexity of an unrestricted variable-length sequence model.

Gomes et al. investigated customer representations based on online shopping sessions for purchase-intention prediction [2]. Their use of embeddings illustrates how session information can be transformed into a predictive representation that captures more than individual clicks. The current implementation uses

engineered structured features rather than learned embeddings. This choice is supported by the tabular nature of the validated implementation and by the desire to preserve interpretable feature groups. The limitation is that fixed engineered features may capture less sequential detail than a learned representation, but they allow direct use of efficient tree models and explicit leakage control.

Esmeli, Bader-El-Den, and Abdullahi studied early purchase-intention prediction in online session-based retailing [3]. Early prediction is important because a system must estimate intent before the outcome if that estimate is to influence recommendation. This work reinforces the need to remove purchase events from model input. If the transaction event itself were retained, the task would no longer represent early prediction. The present project constructs the target from the complete session, removes purchase events, and predicts from the remaining behaviour.

Fabra, Àlvarez, and Ezpeleta examined log-based session profiling and online behavioural prediction in e-commerce websites [4]. Their work demonstrates how logs can be converted into session profiles suitable for prediction. The present preprocessing stage follows the same general principle: raw events are cleaned, ordered, grouped, and summarized. Its distinctive extension is that the resulting purchase probability is not treated as the final application output. It becomes one input to recommendation re-ranking.

A recurring issue in purchase prediction is class imbalance. Purchase sessions commonly form a smaller class than non-purchase sessions, so accuracy alone can conceal poor positive-class recognition. Precision, recall, F1-score, ROC-AUC, PR-AUC, and log loss provide complementary views. The current project also distinguishes probability evaluation from thresholded classification. ROC-AUC, PR-AUC, and log loss use continuous predictions, while precision, recall, and F1 depend on a chosen threshold. The recommendation pipeline retains the continuous probability so that information is not discarded through premature binarization.

2.5 Tree-Based Learning for Structured Session Features

LightGBM is a gradient-boosted decision-tree framework designed for efficient learning from structured data [9]. Ke et al. introduced methods including Gradient-based One-Side Sampling and Exclusive Feature Bundling to improve efficiency. Boosted trees are suitable for session features because relationships between event counts, diversity, duration, and price statistics may be non-linear and interactive. In the implemented ensemble, LightGBM uses 200 estimators, a learning rate of 0.05, 31 leaves, and balanced class weighting.

XGBoost is another boosted-tree learner used in the project. Although the attached paper does not include a separate XGBoost reference, it validates XGBoost as one of the implemented base classifiers. Its configuration uses 200 estimators, a learning rate of 0.05, and maximum depth 5. The controlled depth limits the complexity of individual trees, while boosting combines them to model non-linear decision boundaries.

Random Forest supplies a bagging-based perspective that differs from sequential boosting. It is configured with 300 trees, maximum depth 12, and balanced class weighting. Diversity among model families is useful in a stack because base learners need not make identical errors. Random Forest

recorded the highest accuracy, precision, and F1-score among the individual learners in the reported default-threshold comparison, whereas XGBoost recorded the highest recall. These differences provide a practical reason to combine their probability outputs rather than assuming one learner dominates every metric.

2.6 Stacking and Continuous Intent Estimation

The implemented stacking procedure trains LightGBM, XGBoost, and Random Forest as base learners and uses Logistic Regression as the meta-classifier. Five-fold stratified cross-validation produces out-of-fold probabilities for training the meta-classifier. Stratification preserves the class distribution across folds, which is particularly important for the imbalanced purchase task. Out-of-fold generation ensures that the meta-classifier is trained on predictions for records that were not used to fit the corresponding base learner.

For a new session, fold-specific predictions from each base model are averaged and supplied to Logistic Regression. The meta-classifier then produces the final continuous purchase probability. Logistic Regression is appropriate at this level because the inputs are already probability-like summaries from the base learners; it can learn a stable combination without introducing an unnecessarily complex second-stage model.

The literature reviewed above supports session representation and structured prediction, but the present implementation adds an important application distinction. Binary classification is used only to evaluate whether a session is assigned to the purchase or non-purchase class at a selected threshold. Recommendation uses the full probability. Consequently, sessions with different strengths of intent remain distinguishable even if both fall on the same side of the classification threshold.

2.7 Anomaly Detection and Isolation Forest

Isolation Forest was introduced by Liu, Ting, and Zhou as an anomaly-detection method based on isolating observations through random partitioning [10]. Unusual observations tend to require fewer partitions because they occupy sparse or distinctive regions of the feature space. This approach avoids the need to model a complete normal distribution and can operate without labelled anomaly examples. These properties make it suitable for brand-level behavioural analysis when verified risk labels are unavailable.

The TrustAware Behavioral Suspiciousness Model applies Isolation Forest to seven standardized behavioural features. The validation material confirms the number of features but does not disclose their individual names. The report therefore does not invent or infer feature labels. StandardScaler places feature dimensions on a comparable scale before anomaly analysis. The Isolation Forest uses 100 estimators and a contamination setting of 0.08, which controls the expected proportion used internally by the anomaly process.

The meaning of the output requires restraint. Isolation Forest identifies unusual combinations of input characteristics; it does not provide a causal explanation and does not establish criminal or fraudulent conduct. A brand can be unusual for legitimate reasons. The project therefore interprets model output as behavioural suspiciousness, converts it into trust-related information through the implemented data

pipeline, and uses it only as one influence on product order.

2.8 Behavioural Fraud Research and the Present Distinction

Zhang et al. presented eFraudCom, an e-commerce fraud-detection system based on competitive graph neural networks [11]. Yin et al. investigated behavioural graph fraud detection in e-commerce [12]. These studies demonstrate that interaction networks and behavioural relations can support fraud-oriented analysis. They also highlight what a genuine fraud-detection claim normally requires: a task definition, suitable labels or verified outcomes, a representation of relationships, and evaluation against fraud ground truth.

The present project differs in data, objective, and interpretation. It operates on aggregated brand behaviour from clickstream data, uses an unsupervised Isolation Forest, and has no verified fraud label for each brand. The output is therefore not called fraud probability. The terms behavioural suspiciousness, behavioural trust, and behavioural-risk category are used consistently. This distinction prevents the literature on fraud detection from being used to overstate what the implemented model can conclude.

2.9 Trust-Aware Recommendation

Trust-related information can improve recommendation by adding evidence that is not captured by item relevance alone. Prior recommendation research has investigated user acceptance, hybrid context, popularity effects, and feedback characteristics [13, 14, 5]. The present project applies the underlying principle that ranking can consider more than preference, but defines trust operationally through behavioural consistency rather than social trust or explicit user ratings.

The trust signal is associated with candidates through the product brand. This is structurally different from session intent. Every candidate in the same session shares the session’s purchase probability, but products can have different brands and therefore different trust values and risk categories. This granularity enables trust to change the relative ordering of products that otherwise have similar relevance. It also allows the integration audit to count promotions, demotions, and candidates that require fallback trust.

Soft re-ranking is central to the design. A hard rule would remove all candidates in a selected category, implicitly treating the model output as conclusive. Soft reduction acknowledges uncertainty and retains the candidate for consideration. Low-risk brands experience little or no reduction, Medium-risk brands receive a stronger reduction, and High-risk brands receive the strongest reduction. The exact score equation and parameter derivations are intentionally omitted from this report in favour of a viva-friendly conceptual explanation: relevance, purchase intent, and trust jointly determine priority, while behavioural risk progressively lowers it without mandatory deletion.

2.10 Research Gap

The literature provides mature foundations for each individual part of the problem. Session-aware recommendation captures short-term preference; clickstream models estimate purchase intent; tree ensembles learn from structured features; Isolation Forest identifies unusual observations; and trust research motivates broader ranking criteria. However, these strands are usually evaluated separately.

Purchase prediction commonly ends with a probability or class label, anomaly detection ends with an anomaly score, and recommendation focuses on relevance.

The identified gap is an end-to-end pipeline in which continuous session purchase probability and brand-level behavioural trust are attached to the same candidate records and jointly influence recommendation order. The present project addresses this gap while maintaining strict semantic separation: probability is an intent signal, suspiciousness is an anomaly-derived behavioural signal, trust is a ranking input, and none of these alone is a fraud decision.

2.11 Chapter Summary

The reviewed research supports the use of session data, engineered clickstream features, tree-based learners, stacking, multi-channel retrieval, and unsupervised anomaly analysis. It also establishes the need for careful evaluation and cautious interpretation. The next chapter translates these foundations into system requirements and defines what the implemented project must accomplish as a coherent application.

CHAPTER 3

SYSTEM ANALYSIS AND REQUIREMENTS

3.1 Introduction

System analysis converts the research objective into implementable responsibilities. The system must process raw behavioural data, learn two different kinds of evidence, retrieve products, integrate outputs at the correct granularity, and expose results in a form that can be evaluated. This chapter describes the existing recommendation problem, the proposed integrated system, functional requirements, non-functional requirements, data requirements, and technical feasibility.

3.2 Analysis of the Existing Recommendation Problem

A conventional catalog can return popular products even when no session history is available. As interactions accumulate, co-view and category relationships provide more specific candidates. These mechanisms are useful because they are explainable and efficient, but they primarily measure relevance. If intent prediction is added without brand analysis, the system becomes more responsive to conversion likelihood but still lacks a behavioural trust dimension.

An independent brand anomaly process does not solve this integration problem by itself. Its output must be associated with products through brand identifiers, and missing or insufficient evidence must be handled explicitly. The final ranking must then apply the information in a manner proportionate to its uncertainty. Therefore, the key system-analysis issue is not only whether each model works, but whether its output reaches the correct candidate record and affects ordering in a verifiable way.

3.3 Proposed Integrated System

The final system contains the User Intent Prediction Model, the TrustAware Behavioral Suspiciousness Model, candidate product retrieval, and trust-aware recommendation re-ranking. These are not separate competing applications. They are stages of a single workflow.

The User Intent Prediction Model accepts a fixed session feature vector and returns a continuous purchase probability. The TrustAware Behavioral Suspiciousness Model accepts standardized brand feature records and produces stored suspiciousness, trust, and behavioural-risk information. Candidate retrieval supplies a set of potentially relevant products. Integration joins the session probability by session identifier and trust information by product brand. The final ordering considers all three evidence types and returns the highest-ranked products.

3.4 Functional Requirements

Functional requirements serve as the blueprint of a system, explicitly defining the specific behaviors, features, and operations that the software must execute to satisfy users. They detail the precise interactions between the user and the system, capturing how data is inputted, processed, and subsequently outputted.

Table 3.1. Functional requirements of the integrated system

ID	Requirement	Description
FR1	Load clickstream data	Read the validated event fields required for session, product, category, brand, price, user, and time analysis.
FR2	Clean records	Remove duplicate records, represent missing brands consistently, convert event time, and exclude the category-code field whose missingness prevents dependable use.
FR3	Form sessions	Group by user_session, sort events chronologically, and exclude sessions with insufficient interactions for intent modelling.
FR4	Construct labels safely	Mark sessions containing at least one purchase as positive, then remove purchase events from predictive input.
FR5	Generate session features	Calculate aggregate activity, diversity, price, time, and five recent-action features in a consistent order.
FR6	Predict intent	Load or train the LightGBM, XGBoost, Random Forest, and Logistic Regression stacking pipeline and return continuous probability.
FR7	Evaluate binary intent	Apply the optimized threshold only when classification labels and threshold-dependent metrics are required.
FR8	Generate brand records	Aggregate the validated seven behavioural features for each brand without inventing unavailable feature definitions.
FR9	Standardize brand features	Apply the fitted StandardScaler before Isolation Forest analysis.
FR10	Estimate suspiciousness	Use the 100-estimator Isolation Forest with contamination 0.08 and store the resulting behavioural information.
FR11	Categorize behavioural risk	Associate scored brands with Low, Medium, or High behavioural-risk categories and applicable trust ranges.
FR12	Retrieve candidates	Combine co-view, related-category, and global-popularity channels.
FR13	Consolidate candidates	Remove duplicate session-product pairs, retain the strongest available relevance, and bound source relevance to the range 0.1–1.0.

ID	Requirement	Description
FR14	Integrate model outputs	Join intent by session and trust by product brand; apply documented fallbacks where reliable trust is unavailable.
FR15	Re-rank softly	Increase or reduce ranking priority according to relevance, intent, trust, and risk category without automatically deleting products.
FR16	Produce recommendations	Sort candidates and return the requested top- list with fields required by the application and evaluation.
FR17	Record evaluation data	Compute purchase-prediction, integration, relevance, ranking-quality, average-trust, and exposure metrics.

3.5 Non-Functional Requirements

Correctness is the primary non-functional requirement. Training and inference must use the same session feature order; probability and trust must be joined using the correct keys; and the report must distinguish generated candidate rows from the subset used in the integration audit. Reproducibility requires fixed configurations, stored preprocessing objects, saved model artifacts, and documented evaluation sessions.

Interpretability is also essential. The system should preserve the origin of relevance, intent, and trust so that ranking changes can be inspected. Responsible terminology is part of interpretability: the behavioural component must not be described as determining fraud. Maintainability requires modular separation between preprocessing, model inference, retrieval, integration, and presentation. Efficiency is required to process the validated evaluation set without unnecessary repeated model training. The application should load saved artifacts and reusable lookup structures where appropriate.

Table 3.2. Principal non-functional requirements

Quality	Requirement
Correctness	Prevent target leakage, preserve feature order, use correct integration keys, and report validated values without modification.
Reproducibility	Retain model configurations, preprocessing objects, held-out data boundaries, and saved outputs needed to repeat evaluation.
Interpretability	Expose separate relevance, intent, trust, and risk fields; describe suspiciousness as unusual behaviour rather than confirmed fraud.
Maintainability	Isolate data preparation, intent prediction, brand analysis, retrieval, ranking, API, and interface responsibilities.
Efficiency	Use precomputed relationships and loaded artifacts for evaluation and demonstration rather than retraining on each request.
Usability	Present ordinary product recommendations to customers and detailed model information only in an appropriate evaluation or administrative view.
Data integrity	Handle duplicates, missing brands, event chronology, and insufficient brand evidence consistently.

3.6 Data Requirements

The input must contain event time, event type, product identifier, category identifier, brand, price, user identifier, and user-session identifier. The session identifier and event time are required to reconstruct behaviour. Product, category, and brand fields support diversity measures and candidate association. Price supports session statistics. Event type distinguishes views, cart actions, and purchase outcomes.

The system also requires a product catalog that maps product identifiers to category and brand, candidate-retrieval structures derived from interaction data, saved intent-model artifacts, a fitted StandardScaler, and a brand trust registry. Evaluation requires known purchased products or relevant items for held-out sessions so that HitRate, Precision, Recall, and NDCG can be calculated.

3.7 Input and Output Analysis

The User Intent Prediction Model receives a numerical feature vector and outputs a probability between zero and one. The TrustAware Behavioral Suspiciousness Model receives standardized brand records and yields stored behavioural suspiciousness, trust, and category information. Candidate retrieval outputs session-product pairs with source relevance. Integration produces enriched candidate rows containing the candidate product, associated brand, purchase probability, trust information, and risk category. Re-ranking outputs an ordered list.

The customer-facing output should remain product-oriented. It can include product identifier, brand, category, price, and presentation fields available in the catalog. Detailed suspiciousness and risk information is more appropriate for an administrative or evaluation view because it is an internal

modelling signal and may be misunderstood by customers. This separation also supports responsible communication of unsupervised results.

3.8 Technical Feasibility

The system is technically feasible because the two modelling tasks operate on structured numerical data and use established Python machine-learning libraries. The session task can be trained with LightGBM, XGBoost, scikit-learn Random Forest, and Logistic Regression. The brand task can be implemented with StandardScaler and Isolation Forest. Candidate relations and model outputs can be stored in tabular files and loaded into memory for evaluation.

The existing demonstration architecture uses a FastAPI backend, an in-memory data layer, and a React interface. These technologies are supported by the project documentation, but the report does not claim distributed production deployment. The evaluated results are obtained offline from fixed sessions and candidate records. A local interactive interface can demonstrate event handling and recommendation output without establishing real-world scalability.

3.9 Operational and Ethical Considerations

Operationally, model artifacts and catalog versions must remain aligned. A brand lookup built from one catalog version can produce missing associations if used with another. Session feature generation must be identical during training and demonstration inference. Unknown brands and low-evidence brands require explicit fallback handling so that missingness does not silently become extreme trust or extreme suspicion.

Ethically, anomaly-derived categories should be visible only to authorized evaluators and should not be presented as accusations. A High behavioural-risk category indicates that the available feature combination is unusual under the fitted model. It does not establish intent, illegality, counterfeit status, or customer harm. Human review and corroborating evidence would be required before using the signal for decisions beyond recommendation order.

3.10 Chapter Summary

This chapter defined the project as one integrated system and translated its objective into functional, data, quality, and interpretation requirements. The following chapter details the methodology used to meet those requirements.

CHAPTER 4

METHODOLOGY

4.1 Methodology Overview

The methodology contains an offline preparation path and a recommendation-generation path. Offline processing cleans the raw clickstream, forms labelled sessions, engineers features, trains the User Intent Prediction Model, aggregates brand behaviour, fits the TrustAware Behavioral Suspiciousness Model, and prepares catalog and retrieval data. Recommendation generation accepts an evaluation session, obtains its purchase probability, retrieves products, attaches brand trust information, applies soft trust-aware re-ranking, and selects the top products.

The two learned components were introduced with their model numbers in the abstract because those identifiers appear in the supplied project materials. Throughout the chapters, descriptive names are used. The **User Intent Prediction Model** estimates the probability of purchase from session behaviour. The **TrustAware Behavioral Suspiciousness Model** identifies unusual brand-level feature combinations and produces trust-related information. The components solve different problems and are joined only at the candidate-record level.

4.2 Dataset

The project uses the 2019-Oct.csv e-commerce behaviour dataset from the multi-category store collection [15]. Each event records the time and type of interaction together with product, category, brand, price, user, and session identifiers. The validated product catalog contains 30,223 products and the behavioural analysis covers 765 scored brands.

Table 4.1. Validated dataset and evaluation statistics

Item	Validated value
Raw clickstream file	2019-Oct.csv
Product catalog	30,223 products
Brand catalog / scored-brand coverage	765 brands
Held-out intent-evaluation sessions	1,069
Recommendation evaluation sessions	1,000
Generated recommendation candidate rows	29,571
Average candidates per evaluation session	29.57
Unique products in integrated candidates	4,960
Unique brands in integrated candidates	733
Candidates in the model-integration audit	29,565

The distinction between 29,571 generated rows and 29,565 audited candidates is retained because both values appear in the validated paper. The larger value describes candidate generation for 1,000

sessions, while the integration table reports the subset on which complete model-usage checks were calculated. The report does not merge or alter these counts.

4.3 Data Preprocessing

Duplicate records are removed so that repeated copies of the same event do not inflate activity counts or association statistics. Missing brand values are represented consistently as unknown. The category-code field is excluded because its missing values prevent dependable use in the validated workflow. Event timestamps are converted to datetime values, after which records are sorted by user session and event time.

Chronological ordering is required for recent-action features and for correct duration calculation. Without sorting, the last five actions could be assigned incorrectly and session start or end time could be reversed. The preprocessing stage therefore establishes both data cleanliness and sequential validity before either model is trained.

The same cleaned data supports two different aggregation levels. Session aggregation produces one intent-prediction record per eligible shopping session. Brand aggregation produces one behavioural representation per scored brand. The separation is deliberate: user intent is temporary and session-specific, while behavioural suspiciousness is derived from the wider activity associated with a brand.

4.4 Sessionization and Purchase Labels

Events are grouped by the `user_session` identifier. Within each group, chronological order is preserved. Sessions with too few interactions are excluded because extremely short sequences provide insufficient behavioural context for reliable feature construction. The resulting groups form the basic observations used by the intent model.

A session is assigned the positive purchase label if at least one purchase event is present; otherwise, it receives the non-purchase label. After target creation, purchase events are removed from the model input. Only non-purchase behaviour is used to form predictors. This step prevents direct outcome leakage and ensures that the learning problem remains meaningful: the model estimates the likelihood of purchase from the behaviour that preceded it.

The held-out intent evaluation contains 1,069 sessions. Five-fold stratified cross-validation is used during stacking so that each fold maintains a similar purchase/non-purchase proportion. The held-out data is kept separate for the metrics reported in Chapter 6.

4.5 Session Feature Engineering

The session representation contains 18 validated features. These features combine overall engagement, breadth of exploration, price context, time context, and recent actions. The feature groups are shown in Table 4.2.

Table 4.2. Feature groups used by the User Intent Prediction Model

Group	Features	Purpose
Activity volume	Total events, product views, cart actions	Represents the overall intensity of engagement within the session.
Exploration diversity	Unique products, unique categories, unique brands	Distinguishes narrow examination from broad catalog exploration.
Price context	Average, minimum, and maximum observed product price	Represents the price range and spending context of viewed or carted products.
Temporal context	Session duration, starting hour, day of week, weekend indicator	Captures duration and the time context in which the session occurred.
Recent sequence	Five most recent non-purchase action codes	Preserves limited order information close to the prediction point.

The recent-action encoding assigns 1 to a product view and 2 to a cart action. When fewer than five non-purchase actions are available, unused positions are filled with 0. This fixed-width representation permits tree models to learn whether a recent cart action differs from a sequence dominated by views. Aggregate features remain necessary because the last five events alone do not describe total session duration, diversity, or price range.

Feature engineering uses only information available within the session and before the outcome event. No field derived from the purchase itself is supplied to the model. This preserves the boundary between target creation and predictor construction.

4.6 User Intent Prediction Model

4.6.1 Base Learners

The User Intent Prediction Model uses LightGBM, XGBoost, and Random Forest as base learners. Their validated configurations are summarized in Table 4.3.

Table 4.3. Validated configuration of the intent base learners

Learner	Configuration reported in the project paper
LightGBM	200 estimators; learning rate 0.05; 31 leaves; balanced class weighting.
XGBoost	200 estimators; learning rate 0.05; maximum depth 5.
Random Forest	300 trees; maximum depth 12; balanced class weighting.
Meta-classifier	Logistic Regression trained on out-of-fold base-model probabilities.
Cross-validation	Five-fold stratified cross-validation for stacking predictions.

The learners offer complementary inductive behaviour. LightGBM and XGBoost build additive

boosted trees, while Random Forest averages a collection of bagged trees. This diversity is useful because clickstream features can contain non-linear thresholds and interactions. For example, a cart action may have a different implication in a long, diverse session than in a short, focused session. The ensemble allows different learners to capture such structures and lets the meta-classifier determine how their probabilities should be combined.

4.6.2 Stacking Procedure

Five stratified folds are created from the training data. For each fold, the base learner is trained on four folds and predicts probabilities for the held-out fold. Repeating this process produces out-of-fold probabilities for every training session. These probability columns become the input to Logistic Regression. Because the meta-classifier sees predictions generated for sessions excluded from the corresponding base-model fit, the procedure reduces the risk of learning from optimistically fitted training predictions.

At inference time, fold-specific base learners produce probabilities for a new session. Their predictions are averaged by learner and supplied to the Logistic Regression meta-classifier. The output is a continuous probability expressing the strength of purchase intent. It is stored with the session and joined to every candidate retrieved for that session.

4.6.3 Continuous Probability and Binary Evaluation

The system maintains a strict distinction between probability and class label. The continuous probability is used during recommendation because it retains information about degree. A session with probability 0.88 should not be treated identically to one with probability 0.12 simply because both are below a selected classification threshold.

Binary classification is required only when computing threshold-dependent evaluation metrics. The validated threshold search selected 0.90 because it produced the strongest reported F1-score among the tested values. This high threshold makes positive predictions more selective. It improves precision at the cost of recall, which is appropriate to discuss as an evaluation trade-off. The threshold does not replace or discretize the probability used by the recommendation engine.

4.7 TrustAware Behavioral Suspiciousness Model

4.7.1 Brand-Level Behavioural Representation

The TrustAware Behavioral Suspiciousness Model operates at brand level rather than session level. Clickstream activity is aggregated into seven validated behavioural features for each brand. The attached validation material does not disclose the individual names of those features. Consequently, this report does not invent labels, infer proxy names, or present unsupported calculations. It describes only the confirmed processing sequence and output.

Brand-level aggregation provides a wider behavioural view than a single session. It is intended to determine whether the combination of a brand's observed characteristics is unusual relative to the fitted feature space. Because brands can have different activity levels and feature scales, raw values are not passed directly to Isolation Forest.

4.7.2 Standardization

The seven features are transformed using a fitted `StandardScaler`. Standardization centres and scales each dimension using parameters learned from the training data. This prevents a feature with a numerically large range from dominating distance and split behaviour merely because of its units. The fitted scaler is retained so that later brand records receive the same transformation.

Standardization is a preprocessing operation, not a statement that the features are normally distributed. Its purpose is to present comparable numerical ranges to the anomaly-detection stage. The report specifically uses `StandardScaler` because that is the method validated by the attached paper.

4.7.3 Isolation Forest Analysis

Isolation Forest analyses the standardized records through an ensemble of 100 estimators with contamination set to 0.08. Each estimator repeatedly partitions the feature space. Records that are isolated with relatively short paths exhibit combinations that are easier to separate from the majority and therefore contribute stronger anomaly evidence.

The model is unsupervised. It is not trained on verified examples of fraudulent and non-fraudulent brands. Its result is interpreted as behavioural suspiciousness. The implemented processing converts this information into a trust-related value and stores a behavioural-risk category. The exact internal transformation from raw Isolation Forest output to stored suspiciousness and trust is not specified in the validation material; therefore, no derivation or replacement formula is introduced here.

4.7.4 Trust Estimation and Behavioural-Risk Categories

The scored-brand output contains suspiciousness values from 0.0005 to 0.8500 and trust values from 0.1500 to 0.9995. Brands are placed into Low, Medium, and High behavioural-risk categories. Table 4.4 presents the validated distribution and trust ranges.

Table 4.4. Validated behavioural-risk distribution and trust ranges

Category	Brands	Percentage	Trust range
Low	710	92.81%	0.7000–0.9995
Medium	49	6.41%	0.4000–0.6999
High	6	0.78%	0.1500–0.3999
Total	765	100.00%	0.1500–0.9995

These categories are ranking-oriented behavioural categories. High risk means that the stored behavioural evidence is more unusual and is associated with lower trust under the implemented model. It is not a legal conclusion, verified fraud label, or guarantee that the brand has acted improperly. The category is used to vary ranking priority and to measure exposure in evaluated recommendation lists.

4.8 Candidate Product Retrieval

Candidate generation uses three complementary mechanisms. Co-view retrieval identifies products that have appeared in viewing contexts related to products in the active session. It is the most directly session-linked channel because it uses interaction association. Related-category retrieval expands the candidate set with products from categories connected to the user's present browsing context. Global-popularity retrieval adds widely interacted products and supplies coverage when the session is sparse or when association tables return too few results.

Each channel assigns a source relevance value. A product may be returned by multiple channels. Duplicate session-product pairs are removed, and the highest available relevance is retained so that repeated retrieval does not create repeated recommendations. The final candidate relevance is restricted to the range 0.1–1.0. The lower bound prevents a retrieved candidate from having no effective relevance, while the upper bound ensures that source relevance remains on a controlled scale.

Global popularity is used as a retrieval mechanism and fallback source, not as a separate final recommendation system. Similarly, co-view and category methods are channels within the implemented pipeline rather than independent systems presented to the user. Their outputs are consolidated before model information is attached.

4.9 Trust Integration and Fallback Handling

The continuous purchase probability is joined using the session identifier. Consequently, all candidates from the same evaluation session share the same intent value. Trust information is joined through the candidate product's brand, so candidates in the same session can receive different trust and risk information.

The integration audit reports that all 29,565 evaluated candidates received a real probability from the User Intent Prediction Model. The TrustAware Behavioral Suspiciousness Model provided real trust for 26,309 candidates, equal to 88.99% of the audited set. The remaining 3,256 candidates required fallback trust because ordinary brand evidence was not available.

Unknown-brand candidates numbered 2,460, or 8.32%, and received a neutral trust value of 0.50. A further 796 candidates, or 2.69%, belonged to brands with fewer than five lifetime events and received a trust value of 0.60. These values express limited evidence. They are not claims that unknown or low-activity brands are fraudulent, and fallback behaviour is not treated as a separate recommendation system.

4.10 Trust-Aware Recommendation Ranking

Trust-aware ranking combines three substantive signals and one category-sensitive control. Candidate relevance expresses how strongly retrieval connects the product to the session. Continuous purchase probability expresses the strength of current buying intent. Brand trust expresses the consistency of the brand's aggregate behavioural pattern. The behavioural-risk category determines the strength of the soft ranking reduction.

The process is intentionally described without a ranking equation or parameter derivation. Conceptually, a highly relevant candidate in a session with strong purchase intent can receive increased priority, especially when its associated brand has strong behavioural trust. A candidate linked to greater suspiciousness can receive lower priority. Low-risk brands receive little or no reduction, while Medium- and High-risk categories receive progressively stronger reductions. Products remain in the candidate set; the method changes their order rather than automatically blocking or deleting them.

This approach is suitable for a signal derived from unsupervised analysis. It acknowledges behavioural evidence but avoids treating that evidence as conclusive. It also preserves the ability of relevance to matter: a lower-trust product is not assigned a fraud label, and a higher-trust product is not guaranteed to be relevant. Final priority emerges from the joint consideration of the available signals.

4.11 Evaluation Methodology

Purchase prediction is evaluated with accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and log loss. Accuracy measures the overall fraction of correct thresholded labels. Precision describes the fraction of predicted purchases that are actual purchases, while recall describes the fraction of actual purchases identified. F1-score balances precision and recall. ROC-AUC measures discrimination across thresholds, PR-AUC emphasizes positive-class performance under imbalance, and log loss evaluates the quality of probability estimates.

Recommendation is evaluated at cutoffs 5 and 10. HitRate indicates whether the relevant purchased product occurs in the top- list. Precision measures the fraction of recommended positions that are relevant. Recall measures the proportion of relevant items retrieved; under the validated evaluation setup with one target product per session, HitRate and Recall have the same reported values. NDCG rewards relevant items appearing nearer the top. Average Trust summarizes the trust composition of recommended products, and High-Risk Exposure measures the proportion of recommended positions occupied by High behavioural-risk brands.

Integration measures verify that the system actually uses model outputs. They count real probability usage, real trust usage, fallback usage, positions displaced by trust, high-trust items promoted into the top three, and Medium/High-risk items demoted. These checks are necessary because a pipeline can appear to contain two models while silently relying on default values during ranking.

4.12 Chapter Summary

The methodology transforms raw clickstream events into two distinct learned signals, retrieves candidates through three channels, joins evidence at session and brand levels, and applies conceptual soft re-ranking. Detailed TrustAware equations and score calculations are intentionally excluded. The next chapter explains how the methodology is organized within the implemented system.

CHAPTER 5

SYSTEM DESIGN AND IMPLEMENTATION

5.1 Overall System Architecture

The supplied architecture image in Figure 5.1 is the only project architecture or workflow figure included in this report. It shows the progression from raw clickstream data through filtering and sessionization, parallel intent and brand analysis, candidate retrieval, and trust-aware re-ranking and evaluation.

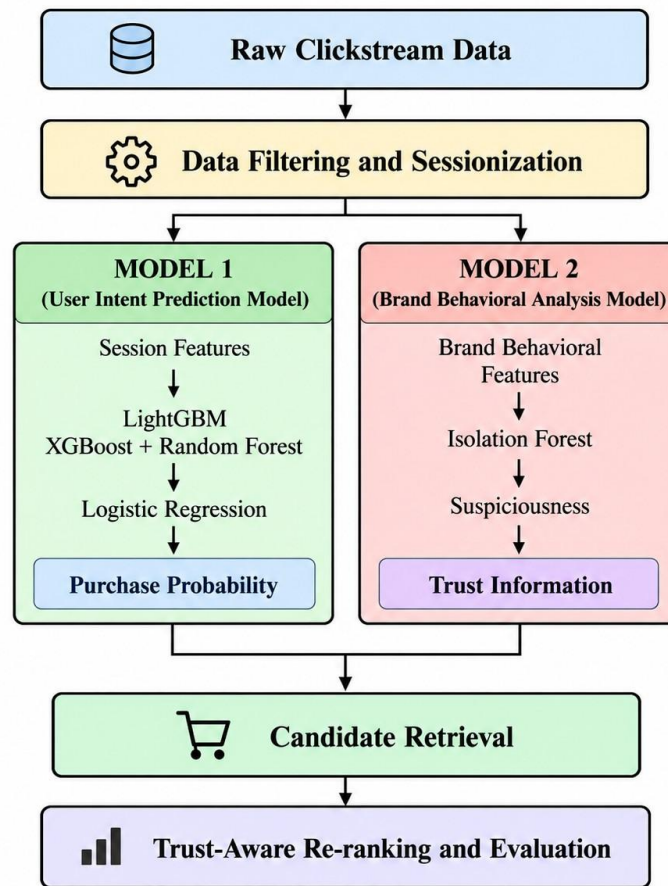


Figure 5.1. Overall system architecture supplied with the project

The parallel branches reflect the different units of analysis. Session features are processed by the User Intent Prediction Model to obtain purchase probability. Brand behavioural features are processed by the TrustAware Behavioral Suspiciousness Model to obtain suspiciousness and trust information. The outputs meet at candidate retrieval and integration. The architecture does not imply that suspiciousness is shown directly to customers or that the brand component makes a fraud decision.

5.2 Component Organization

The implementation is divided into data preparation, intent modelling, brand analysis, candidate retrieval, recommendation integration, application access, and evaluation. Modular organization reduces coupling. A change in candidate retrieval should not require the intent model to be retrained, while a refreshed brand registry can be loaded without changing session feature construction.

Table 5.1. Responsibilities of the implemented components

Component	Primary responsibility
Preprocessing	Clean raw events, normalize fields, order interactions, and produce model-ready aggregates.
Session feature builder	Generate the same 18 features for training and later inference.
User Intent Prediction Model	Produce continuous purchase probability from the stacked base learners and Logistic Regression.
Brand feature builder	Produce the seven validated brand-level feature columns used by the behavioural model.
TrustAware Behavioral Suspiciousness Model	Standardize brand data, apply Isolation Forest, and prepare stored suspiciousness, trust, and risk information.
Candidate retrieval	Gather co-view, related-category, and popular products and consolidate duplicate records.
Integration and ranking	Join session and brand outputs, apply documented fallbacks, and order candidates through soft trust-aware prioritization.
Backend	Load artifacts, accept session/event requests in the demonstration application, and return catalog or recommendation data.
Customer interface	Display ordinary product information and recommendations without presenting internal risk judgements as customer claims.
Administrative evaluation view	Display session intent, trust coverage, ranking effects, and validated metrics for inspection.

5.3 Data-Preparation Implementation

The preprocessing implementation reads the required columns and applies type conversion before aggregation. Duplicate removal occurs before event counting so that repeated input rows do not inflate session intensity. Missing brands are mapped to a consistent unknown representation. Sorting is performed by session and event timestamp, establishing deterministic recent-action order.

Session aggregation produces one row per eligible session. Purchase labels are computed from the original event sequence, after which purchase events are excluded from feature input. This order is critical: removing purchase events before label generation would erase the outcome, while retaining them after label generation would leak it. The implementation therefore separates label extraction from

feature extraction.

Brand aggregation is maintained as a distinct procedure. The resulting seven features are passed through the fitted `StandardScaler`. Since feature names are not available in the validation material, the implementation documentation should preserve the trained column order and scaler metadata rather than attempting to relabel the features in the report.

5.4 User Intent Prediction Implementation

The base learners are trained within five stratified folds. For each model family, fold-specific fitted objects are retained for inference. Their held-out predictions form the meta-training matrix for Logistic Regression. Saved artifacts include the fitted base models, the meta-classifier, and metadata describing the feature order and selected threshold.

During inference, a session feature row is validated against the expected 18-feature schema. Each fold-specific model returns a probability. Probabilities are averaged for each base-learner family, arranged in the same order used during meta-training, and passed to Logistic Regression. The resulting continuous value is the purchase-intent output consumed by recommendation.

The 0.90 threshold is stored for classification reporting and dashboard labelling. It is not applied before ranking. This prevents the implementation from losing probability resolution. The recommendation system can therefore respond gradually to changes in session behaviour even when the binary label remains unchanged.

5.5 TrustAware Behavioral Suspiciousness Implementation

The behavioural implementation loads the brand feature table, applies the fitted `StandardScaler`, and invokes the 100-estimator Isolation Forest configured with contamination 0.08. Resulting anomaly evidence is processed into stored suspiciousness and trust fields and associated with the validated risk categories. The generated registry uses brand as the lookup key for candidate integration.

No detailed trust equation is included because the authoritative validation material does not specify the transformation. This is an implementation-boundary decision rather than an omission to be filled with an invented formula. The report documents the input, confirmed preprocessing, model configuration, output ranges, category distribution, and integration behaviour—the information required to explain and evaluate the component accurately.

The registry contains 765 scored brands. When a candidate brand is absent, the integrator identifies whether the brand is unknown or has insufficient lifetime evidence and applies the corresponding fallback. The source of trust should be retained as metadata so that evaluation can distinguish model-derived values from fallbacks.

5.6 Candidate Retrieval Implementation

The retrieval layer is designed as a union of three result sources. Co-view retrieval consults product association information derived from session activity. Related-category retrieval uses the active product

or category context to obtain additional items. Popularity retrieval contributes catalog coverage. Each returned record includes the candidate product, source, and relevance.

Candidates are grouped by session and product. When multiple sources return the same product, only one record is kept and the maximum relevance is used. Source provenance may be retained for debugging, but duplication does not create multiple recommendation positions. Values are clipped to the validated 0.1–1.0 interval before integration.

The retrieval layer intentionally precedes trust-aware ranking. It seeks products that are potentially relevant and does not suppress candidates according to behavioural category. This separation enables the system to apply a soft ranking response to a common candidate pool and makes the influence of trust measurable through position changes.

5.7 Recommendation Integration and Re-Ranking Implementation

For each candidate row, the integration layer retrieves the session probability and brand trust record. Validation checks ensure that the probability lies within the expected range and that trust and suspiciousness fields are consistently represented. Candidate rows with model-derived trust are marked separately from those using unknown-brand or low-evidence fallbacks.

The ranking implementation then combines relevance, intent, trust, and category-level reduction. It does not convert purchase probability to a binary decision. It does not discard candidates solely because they are associated with the High behavioural-risk category. Instead, it applies the strongest reduction to that category, a smaller reduction to Medium risk, and little or no reduction to Low risk. Candidates are sorted by resulting priority, and the requested top- records are returned.

Tie handling should be deterministic, using stable secondary fields such as relevance or product identifier where required. Determinism supports reproducibility: the same session, catalog, model artifacts, and retrieval data should yield the same ordered result.

5.8 Backend and Application Integration

The project documentation describes a FastAPI backend with an in-memory data layer and a React-based demonstration interface. At startup, the backend loads the product catalog, saved intent models, metadata, and brand trust registry. The in-memory design simplifies local evaluation and avoids dependence on an external database. It is appropriate for a final-year demonstration but does not constitute a claim of distributed or production-scale persistence.

The event-handling path associates activity with a session and rebuilds the required session representation. Recommendation access retrieves candidates and returns an ordered product list. Catalog and search operations support normal store interaction. The evaluation path exposes the evidence required to inspect model usage and ranking influence.

The demonstration may update recommendations interactively, but the reported experiment remains an offline evaluation over fixed sessions and candidates. This distinction prevents a local interface from being described as a validated real-time production deployment.

5.9 Customer and Administrative Views

The customer view presents product cards and standard catalog information. Internal behavioural suspiciousness is not presented as a customer-facing accusation. This reduces the risk that an unsupervised score will be interpreted as a verified statement about a brand.

The administrative or evaluation view can present the session probability, thresholded intent label, candidate relevance, source of trust, behavioural-risk category, and rank movement. It can also report aggregate metrics. Such visibility is useful for testing because it confirms whether the two models are connected correctly. It also supports viva explanation: the evaluator can trace how session behaviour leads to purchase probability and how brand behaviour changes product priority.

5.10 Artifact and State Management

The implementation depends on consistent versions of several artifacts: the product catalog, retrieval data, base-model folds, Logistic Regression meta-classifier, feature metadata, StandardScaler, Isolation Forest, and brand trust registry. Version mismatch can cause missing products, reordered feature columns, or invalid brand joins. Artifacts should therefore be generated together and loaded with validation checks.

Active session state in the demonstration includes recent events and the values needed to reconstruct features. Because the data layer is in memory, process restart clears transient state unless it is reconstructed. This limitation is acceptable for local demonstration but should be addressed through persistent storage in a future deployment.

5.11 Testing and Verification

Testing covers preprocessing, leakage prevention, feature shape, model inference, value ranges, retrieval duplication, integration coverage, fallback routing, and ranking determinism. A positive session should retain its label after purchase events are removed from features. A feature row should contain exactly 18 fields in the trained order. All probabilities and stored trust values should remain within their validated ranges.

Candidate testing should confirm that duplicate session-product records are consolidated and that relevance remains between 0.1 and 1.0. Integration testing should verify that every evaluated candidate receives session probability and either model-derived or documented fallback trust. Ranking testing should demonstrate that stronger trust can promote a product and higher behavioural risk can demote it without forcing deletion.

The paper’s integration audit provides empirical verification beyond unit-level checks. It reports 100% real probability usage, 88.99% real trust usage, 2,972 displaced positions, 628 high-trust promotions into the top three, and 1,623 Medium/High-risk demotions. These values show that the final ordering is not merely a relevance list with unused model columns.

5.12 Chapter Summary

The implemented design preserves modularity, correct data granularity, model-output traceability, and cautious interpretation. The supplied architecture figure captures the complete flow, while the textual implementation details explain how model artifacts, retrieval records, fallbacks, backend access, and evaluation fit together. Chapter 6 presents the validated numerical results.

CHAPTER 6

RESULTS AND EVALUATION

6.1 Evaluation Overview

Evaluation covers the User Intent Prediction Model, the TrustAware Behavioral Suspiciousness Model, model integration, and the final recommendation ranking. Numerical values are reproduced from the attached research paper without modification. Reference rankings are included only to contextualize evaluation; they are not described as separate final system implementations.

6.2 User Intent Prediction Results

Four predictive outputs were evaluated on the held-out set: LightGBM, XGBoost, Random Forest, and the final stacking ensemble. Table 6.1 reports threshold-dependent classification metrics under the reported comparison setting.

Table 6.1. Purchase-prediction accuracy and classification metrics

Model	Accuracy	Precision	Recall	F1
LightGBM	0.8483	0.2470	0.5125	0.3333
XGBoost	0.8316	0.2316	0.5500	0.3259
Random Forest	0.8945	0.3300	0.4125	0.3667
Stacking Ensemble	0.7946	0.1966	0.5750	0.2930

Random Forest records the highest accuracy, precision, and F1-score in this comparison, while XGBoost records the highest recall among the individual learners. The stacking ensemble has the highest recall overall in the table but lower precision and accuracy. These differences illustrate why a single threshold-dependent metric is insufficient for model assessment.

Table 6.2. Purchase-prediction probability metrics

Model	ROC-AUC	PR-AUC	Log Loss
LightGBM	0.7812	0.3695	0.3580
XGBoost	0.7791	0.3494	0.4108
Random Forest	0.7814	0.3664	0.3342
Stacking Ensemble	0.7845	0.3861	0.5244

The stacking ensemble achieves the highest ROC-AUC, 0.7845, and the highest PR-AUC, 0.3861. Random Forest achieves the lowest log loss, 0.3342. The ensemble’s discrimination measures indicate that its continuous scores provide the strongest reported ordering of positive and negative sessions, while the higher log loss indicates that probability quality is not uniformly best under every criterion. This

balanced interpretation avoids claiming that the ensemble dominates all models.

6.3 Threshold Optimization

The selected threshold of 0.90 is used for the final binary classification report. At this threshold, accuracy is 0.9371, precision is 0.6364, recall is 0.3500, and F1-score is 0.4516.

Table 6.3. Optimized binary purchase-classification result

Threshold	Accuracy	Precision	Recall	F1-score
0.90	0.9371	0.6364	0.3500	0.4516

The high threshold makes the classifier conservative: fewer sessions are labelled as purchases, increasing precision while reducing recall. This trade-off matters for classification reporting but does not restrict recommendation. The continuous stacking probability remains available and is used directly during ranking.

6.4 Behavioural Suspiciousness and Trust Results

The TrustAware Behavioral Suspiciousness Model scored 765 brands. Suspiciousness ranges from 0.0005 to 0.8500, while trust ranges from 0.1500 to 0.9995. The distribution is repeated in Table 6.4 to emphasize the result rather than the configuration.

Table 6.4. Behavioural suspiciousness and trust results

Behavioural-risk category	Brands	Share	Trust interval
Low	710	92.81%	0.7000–0.9995
Medium	49	6.41%	0.4000–0.6999
High	6	0.78%	0.1500–0.3999

Most scored brands fall in the Low category, while six brands fall in the High category. This distribution is an output of the model and implemented categorization, not a prevalence estimate for fraud. The appropriate conclusion is that six brands display the model’s strongest unusual behavioural characteristics within the evaluated data.

6.5 Model Integration Results

Table 6.5 reports whether real model outputs reached the candidate records and whether trust altered ranking positions.

Table 6.5. Model integration and ranking influence

Integration measure	Validated result
Real User Intent Prediction Model probability usage	29,565 / 29,565 (100.00%)
Fallback intent usage	0.00%
Real TrustAware behavioural trust usage	26,309 / 29,565 (88.99%)
Fallback trust usage	3,256 / 29,565 (11.01%)
Unknown-brand fallback	2,460 candidates (8.32%)
Low-confidence-brand fallback	796 candidates (2.69%)
Candidate rank positions displaced by trust	2,972
High-trust items promoted into top three	628 instances
Medium/High-risk items demoted	1,623 instances

The 100% probability-usage result confirms that the continuous intent output was available for every audited candidate. The 88.99% real-trust coverage shows that the majority of candidates used model-derived brand information. Fallback use is concentrated in identifiable uncertainty cases rather than silently replacing the behavioural model.

The position-change counts demonstrate active influence. Trust displaced 2,972 candidate positions, promoted 628 high-trust items into the top three, and demoted 1,623 Medium/High-risk items. These effects align with the intended behaviour: trust modifies priority without requiring blanket candidate deletion.

6.6 Recommendation Results

Table 6.6 reports the final trust-aware results at cutoffs 5 and 10.

Table 6.6. Final trust-aware recommendation results

Metric	Top 5	Top 10
HitRate	0.8220	0.9410
Precision	0.1672	0.0956
Recall	0.8220	0.9410
NDCG	0.5685	0.6078
Average Trust	0.6677	0.6654
High-Risk Exposure	0.00%	0.00%

HitRate@10 and Recall@10 reach 0.9410, showing that the relevant target occurs in the top ten for 94.10% of evaluated sessions under the validated setup. Precision@10 is lower than Precision@5 because the denominator includes twice as many recommendation positions. NDCG@10 of 0.6078 indicates that relevant items are not merely present but are meaningfully concentrated toward higher positions.

Average Trust remains similar at the two cutoffs, with 0.6677 at five and 0.6654 at ten. High-Risk Exposure is 0.00% at both cutoffs. This does not imply that every possible future session will have zero exposure. It describes the evaluated lists produced from the validated sessions, candidates, model artifacts, and behavioral registry.

6.7 Reference Baseline Context

The paper evaluates reference ranking strategies to determine whether the final integrated method improves ranking quality and behavioural composition. These strategies are retained only as evaluation baselines. They are not additional final systems or separate implementation deliverables.

Table 6.7. Top-ten evaluation context from the project paper

Metric	Popularity	Purchase-only	Hard-filter	Trust-aware
HitRate@10	0.2050	0.9210	0.9060	0.9410
Precision@10	0.0207	0.0937	0.0922	0.0956
Recall@10	0.2050	0.9210	0.9060	0.9410
NDCG@10	0.1107	0.5349	0.5269	0.6078
Average Trust@10	0.5752	0.5999	0.6062	0.6654
High-Risk Exposure@10	0.98%	1.21%	0.00%	0.00%

The final method records the strongest value in every reported Top-10 utility and trust column. NDCG@10 rises from 0.5349 in the purchase-only reference to 0.6078, while Average Trust@10 rises from 0.5999 to 0.6654. High-Risk Exposure@10 is reduced from 1.21% to 0.00%. The hard-filter reference also produces zero exposure but has lower HitRate@10 and NDCG@10 than the final method. This provides evaluation evidence in favour of soft re-ranking: behavioural risk can influence order without requiring unconditional removal.

6.8 Interpretation of Recommendation Quality

HitRate and Recall show strong target coverage in the top ten. Precision appears numerically smaller because only a small number of items are relevant for each evaluated session and the recommendation list contains up to ten positions. NDCG is therefore particularly useful because it distinguishes a relevant item at the first position from the same item near the bottom of the list.

The increase in NDCG indicates that trust-aware prioritization did not merely preserve relevant products somewhere in the list. It improved their discounted positions in aggregate. This result can occur because trust is not applied independently of relevance: the pipeline starts from relevant candidates and then adjusts their priority. Strong trust can help an already relevant item rise, while lower trust can move another item down.

The result should not be generalized beyond the evaluation data without further validation. Offline relevance is derived from observed session outcomes and does not measure customer satisfaction, revenue, long-term retention, or causal response to displayed recommendations. Online experiments would be

required to establish those effects.

6.9 Interpretation of Trust and Exposure

Average Trust describes the composition of the recommendation list under the project-specific trust definition. An increase means that recommended positions are, on average, associated with brands having stronger behavioural trust in the stored registry. It does not mean that a recommendation list has a 66.54% probability of being safe, nor does it measure product quality.

High-Risk Exposure counts recommendation positions associated with the High behavioural-risk category. Zero exposure in the evaluated top ten means that High-category products did not occupy those positions after re-ranking. Because the method does not delete candidates, the same products may remain lower in the candidate ordering. The result therefore reflects prioritization, not universal blocking.

Fallback trust also affects interpretation. Real trust covered 88.99% of audited candidates, while 11.01% used documented fallbacks. The fallback values communicate uncertainty rather than suspiciousness. Future work should increase model coverage through improved brand resolution and additional evidence.

6.10 Purchase-Intent Discussion

The stacking ensemble's highest ROC-AUC and PR-AUC support its use as the source of continuous intent, even though Random Forest performs better on accuracy, precision, F1, and log loss in the default comparison. Ranking benefits from a continuous discriminative signal and does not depend solely on the optimized binary label. This explains why the selected final component can be justified without claiming superiority on every classification metric.

The 0.90 threshold result further shows the effect of class imbalance. Precision increases to 0.6364, but recall falls to 0.3500. In a dashboard, this threshold can label only the most confident sessions as purchase-oriented. In recommendation, retaining the continuous probability allows lower-probability sessions to contribute proportionately rather than being reduced to a single negative class.

6.11 Behavioural-Model Discussion

The behavioural model provides broad scored-brand coverage, with 765 brands and model-derived trust for 88.99% of audited candidates. The fact that 92.81% of brands are categorized as Low risk suggests that the strongest reductions are applied to a small subset. Medium and High categories together account for 55 brands. This distribution limits the extent to which the model can dominate every candidate list while still permitting noticeable rank changes where behavioural evidence differs.

The model configuration is modest and explainable: seven standardized features, StandardScaler, and a 100-estimator Isolation Forest. Its limitation is not computational complexity but semantic ground truth. Without verified outcomes, suspiciousness cannot be calibrated as fraud probability. The current use as a soft ranking signal is therefore more defensible than an automatic enforcement decision.

6.12 Limitations of the Evaluation

The first limitation is that the experiment is offline. Observed purchase events supply relevance targets, but the system is not evaluated through a live controlled experiment. The metrics cannot establish whether customers would click, purchase, or trust the displayed recommendations in production.

The second limitation is the available behavioural evidence. Seven features are validated, but their names are not provided in the available material, limiting detailed feature-level interpretation. The transformation from raw anomaly output to stored trust is also not specified, so the report appropriately avoids a mathematical derivation.

The third limitation is incomplete brand coverage. Unknown brands and brands with fewer than five lifetime events require fallback trust. Although fallback usage is measured and is not dominant, it remains a source of uncertainty. The fourth limitation is temporal and catalog dependence. Trust and popularity can change as new events arrive, and saved values require periodic refresh.

The fifth limitation concerns the evaluation baselines. They provide useful reference points but do not exhaust the space of recommendation algorithms. Neural session models, graph recommenders, and learning-to-rank methods are outside the implemented comparison. The final result should therefore be interpreted as evidence for the proposed integration within the validated design, not as proof of universal superiority.

6.13 Chapter Summary

The results validate all major parts of the pipeline. The User Intent Prediction Model produces a discriminative continuous probability; the TrustAware Behavioral Suspiciousness Model covers most evaluated candidates; integration changes thousands of positions; and the final ranking achieves HitRate@10 of 0.9410, NDCG@10 of 0.6078, Average Trust@10 of 0.6654, and zero High-Risk Exposure@10. These conclusions remain bounded by the offline setting and the non-fraud interpretation of suspiciousness.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Project Summary

This project developed an integrated trust-aware e-commerce recommendation system from clickstream data. The system does not treat recommendation, purchase prediction, and brand analysis as independent outputs. It connects them through a candidate-level pipeline in which current-session intent and brand-level behavioural trust jointly influence the order of relevant products.

The User Intent Prediction Model summarizes sessions through 18 aggregate, price, temporal, diversity, and recent-action features. Purchase labels are constructed from transaction outcomes, but purchase events are removed from model input to prevent leakage. LightGBM, XGBoost, and Random Forest generate base probabilities, and Logistic Regression combines out-of-fold predictions produced by five-fold stratified cross-validation. The model retains continuous purchase probability for recommendation and uses the optimized 0.90 threshold only for binary evaluation.

The TrustAware Behavioral Suspiciousness Model analyses seven validated brand-level features. StandardScaler prepares the features, and a 100-estimator Isolation Forest with contamination 0.08 identifies unusual combinations. The resulting output is processed into behavioural suspiciousness, trust information, and Low, Medium, or High behavioural-risk categories. The component is not a fraud detector. It provides an anomaly-derived signal whose appropriate use is cautious ranking influence.

Candidate retrieval combines co-view, related-category, and global-popularity evidence. Duplicate session-product pairs are consolidated and source relevance is bounded between 0.1 and 1.0. The session probability and brand trust record are attached to every candidate. The final ordering conceptually combines relevance, intent, and trust and applies progressively stronger soft reductions for higher behavioural-risk categories. Products are re-ranked rather than automatically removed.

7.2 Key Findings

The intent evaluation shows that the stacking ensemble achieves ROC-AUC of 0.7845 and PR-AUC of 0.3861, the strongest reported values for those metrics. At the optimized binary threshold of 0.90, the model achieves accuracy of 0.9371, precision of 0.6364, recall of 0.3500, and F1-score of 0.4516. These results demonstrate why probability and class label must remain separate: the ranking can use continuous discrimination even when a selective classification threshold is preferred for reporting.

The behavioural analysis scores 765 brands: 710 Low, 49 Medium, and 6 High behavioural risk. Real trust information covers 88.99% of audited candidates. The trust signal displaces 2,972 positions, promotes 628 high-trust items into the top three, and demotes 1,623 Medium/High-risk items. These counts provide direct evidence that the behavioural component affects recommendation rather than existing only as an unused analytical output.

The final recommendation method achieves HitRate@10 of 0.9410, Precision@10 of 0.0956,

Recall@10 of 0.9410, NDCG@10 of 0.6078, Average Trust@10 of 0.6654, and High-Risk Exposure@10 of 0.00%. Within the validated evaluation, the method improves ranking quality and trust composition while retaining a soft response to behavioural risk.

7.3 Project Contributions

The first contribution is an end-to-end implementation that maintains correct semantic boundaries among session intent, product relevance, and brand trust. The second is leakage-aware feature construction for purchase prediction. The third is a stacking implementation that preserves probability for downstream use. The fourth is a responsibly framed behavioural suspiciousness model that avoids unsupported fraud claims. The fifth is multi-channel candidate retrieval followed by trust-aware soft re-ranking. The sixth is an integration audit that measures actual model usage and rank movement rather than relying only on final aggregate metrics.

7.4 Limitations

The project is limited by the observational clickstream dataset and the absence of verified fraud labels. Behavioural suspiciousness can identify unusual patterns but cannot determine motive or wrongdoing. The seven brand features and the exact trust transformation are not fully described in the available validation material, restricting deeper feature-level explanation. Unknown and low-evidence brands require fallback values.

The evaluation is offline and uses historical sessions. It does not measure live user satisfaction, latency under production load, revenue, fairness across catalog segments, or long-term behaviour. The demonstration backend uses in-memory state and is not a distributed persistence layer. Candidate retrieval is based on co-view, category, and popularity mechanisms rather than a learned end-to-end ranker. These constraints define appropriate boundaries for the conclusions.

7.5 Future Enhancements

7.5.1 Temporal Behavioural Modelling

Future work can analyse how brand behaviour changes over time rather than relying only on an aggregate record. A rolling or windowed representation could distinguish persistent anomalies from short-lived promotions, seasonal demand, or catalog changes. Temporal stability checks could reduce false suspiciousness and indicate when a stored trust value should be refreshed.

7.5.2 Additional Evidence Sources

The trust component could be strengthened through product reviews, return patterns, seller history, price changes, delivery information, verified transaction outcomes, and human-review decisions. Such signals should be integrated only when their meaning and coverage are understood. They should not automatically penalize low-volume brands or brands with missing data.

7.5.3 Ground-Truth Validation and Calibration

Verified platform outcomes could support calibration of behavioural suspiciousness and estimation of whether risk categories correspond to meaningful operational concerns. Human review could examine High-category cases to identify legitimate explanations. Until such validation is available, the current terminology and soft-ranking use should be retained.

7.5.4 Advanced Recommendation Models

Future versions could compare learning-to-rank, neural session, graph-based, or hybrid recommendation models while retaining trust as an explicit objective or feature. Any extension should preserve the distinction between retrieval relevance and behavioural trust and should be evaluated for both recommendation utility and exposure.

7.5.5 Improved Coverage and Cold-Start Handling

Brand resolution can be improved by cleaning aliases, normalizing identifiers, and connecting catalog metadata more consistently. Low-evidence brands could use category-aware uncertainty estimates rather than a single fallback. New-product retrieval could incorporate content information where available, reducing dependence on global popularity.

7.5.6 Production-Oriented Serving

A production extension would require persistent session storage, model and catalog versioning, low-latency retrieval indexes, monitoring, and periodic behavioural-model updates. Online experimentation would be necessary to measure customer response and operational impact. The present FastAPI and React demonstration can serve as a prototype interface, but production claims should be made only after such infrastructure is implemented and tested.

7.5.7 Explainability and Governance

An administrative explanation can show which retrieval channel produced a product, the strength of session intent, whether trust was model-derived or a fallback, and whether behavioural risk changed its position. Access controls and audit logs should govern who can view brand risk information. Customer-facing explanations should remain neutral and should not expose anomaly categories as accusations.

7.6 Final Conclusion

The project demonstrates that session purchase intent and brand-level behavioural trust can be incorporated into one recommendation pipeline without conflating their meanings. The User Intent Prediction Model estimates continuous purchase probability from leakage-controlled clickstream features. The TrustAware Behavioral Suspiciousness Model identifies unusual brand-level patterns and provides trust-related information. Multi-channel retrieval supplies relevant products, and soft re-ranking changes their priority according to relevance, intent, and trust.

The validated results show strong top-ten target coverage, improved ranking quality, increased average trust, and zero High-Risk Exposure in the evaluated lists. The contribution is therefore not a fraud-detection claim. It is a technically integrated and responsibly interpreted method for allowing behavioural suspiciousness to influence e-commerce recommendation order while retaining products and preserving recommendation utility.

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Trust Aware E-Commerce Recommendation Using Brand Risk Analysis

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Abstract—E-commerce platforms generate a continuous stream of user activity through actions such as viewing products, adding to the cart, and purchasing items. Such user activities can provide useful information of the buying intention of the customer during the shopping session. However, using purchase intention for recommending products risks boosting products associated with brands with abnormal behavioral patterns. This paper proposes a trust-aware recommendation method that integrates session-level purchase prediction and brand-level behavioral suspiciousness information.

The purchase prediction module uses aggregated behavioral, price, and time-related features in combination with the user's non-purchase activity during the session. LightGBM, XGBoost, and Random Forest, as base classifiers, and Logistic Regression as the meta-classifier are employed for stacking-based continuous purchase probability estimation. Meanwhile, the brand module uses seven standardized behavioral features and Isolation Forest for identifying unusual patterns to extract suspiciousness and trust information. Both sets of information are integrated into the recommendation module, in which candidate products are gathered using co-view, co-category, and popularity-based retrieval methods.

The retrieved products are re-ranked using candidate relevance, purchase probability, brand trust, and risk penalty. In the experiments, all the candidates have valid Model 1 purchase probabilities while 88.99% have model-derived trust values from Model 2. Given the 1,000 test sessions, the system recommends 29,571 candidate records. The trust-aware ranking increases NDCG@10 from 0.5349 for the purchase-only approach to 0.6078 and results in 0.00% High-Risk Exposure@10. It suggests that behavioral trust information can be leveraged in the recommendation ranking while keeping anomaly detection decoupled from the final fraud decision.

Index Terms—E-commerce recommendation, clickstream analysis, purchase prediction, LightGBM, Isolation Forest, behavioral suspiciousness, trust-aware ranking.

1. INTRODUCTION

E-commerce platforms produce large amounts of data from user interactions during online shopping. During a single shopping session, a customer might browse through several products, switching between different categories, add items to their cart, and abandon the purchase or complete the purchase. All actions that occur during a shopping session are captured as clickstream data. As the events are part of the same shopping session, they can be used to estimate the customer's current purchase intention even in the absence of detailed user information. Prior work has shown that session activity, recent actions, and temporal characteristics can provide valuable information for predicting purchase intention based on clickstream data [1]–[5].

Recommendation systems use customer interaction data to identify products that might be relevant for the current user. Various approaches employ product popularity, item co-occurrence, recent interaction, and session-based models to recommend similar items [6]–[8]. However, such strategies focus predominantly on product relevance and user preferences. Relevance alone does not provide information about the behavioral trust associated with the brand of a recommended product.

This work combines purchase intention and behavioral trust in a unified recommendation pipeline. The first component estimates the purchase probability of the current shopping session. The second component analyzes brand-level behavioral patterns and identifies behavior that is unusual compared with other brands. The output of the second model is represented as behavioral suspiciousness and converted into trust-related information for use during recommendation ranking. This signal is not interpreted as evidence that a brand has committed fraud.

A. Problem Statement

A recommendation system based exclusively on purchase probability, may rank a product as important due to its relevance to the current session, without taking into account the behavioral characteristics associated with its brand. Meanwhile, an independent anomaly detection system may detect anomalous behavior from brands without influencing the products being presented to the customer. Thus, the problem addressed in this work is to develop a recommendation pipeline that jointly considers the purchase intention of the current shopping session and the behavioral trust associated with the brands of candidate products.

B. Limitations of Existing Approaches

Existing session-based recommendation methods focus on retrieving products matching the user's immediate interests and predict relevant items based on current interactions [6]–[8]. Similarly, purchase-intention prediction methods focus on determining if a shopping session is likely to result in a purchase [1]–[4]. Separate research has investigated strange or possibly fraudulent behavior using anomaly detection and graph-based methods [11], [12]. These approaches address different aspects of e-commerce behavior in isolation. A recommender system may therefore provide relevant products without considering available behavioral trust information,

while an anomaly detection model detects unusual behavior without influencing the recommendation list.

C. Proposed Solution

The proposed system combines purchase prediction, brand behavioral analysis and recommendation ranking into a single pipeline. Firstly, the raw clickstream data is cleaned and organized into user sessions. Features describing overall session behavior and recent user actions are then generated for the purchase prediction model. LightGBM, XGBoost and Random Forest are used as base models in a stacking ensemble with Logistic Regression used as the meta-model to combine their outputs. This results in a continuous value representing the purchase probability of the current session.

In parallel, brand-level behavioral features are standardized and processed using Isolation Forest. The model identifies brands with unusual combinations of behavioral characteristics and generates suspiciousness and trust information. The recommendation module retrieves candidate products through session co-view, related-category and global popularity-based channels. The retrieved candidates are then combined with the purchase probability and brand trust information for final ranking.

The final ranking score takes into account candidate relevance, purchase probability, brand trust and a risk-level penalty. Rather than completely removing products associated with higher-risk behavioral categories, the proposed system applies a soft penalty which lowers their ranking position while allowing them to still appear in the candidate pool.

D. Main Contributions

The main contributions of the implemented system are as follows:

- 1) A session-based purchase prediction pipeline that combines aggregate session features and recent user actions using a stacking ensemble.
- 2) A brand-level behavioral suspiciousness model using standardized behavioral features and Isolation Forest.
- 3) Integration of purchase probability and brand trust information directly into the recommendation ranking process.
- 4) Evaluation of the proposed ranking approach against popularity-based, purchase-only, and hard-filter recommendation approaches.
- 5) Validation of model integration, ranking changes, fallback handling, reproducibility, and data-leakage controls.

2. LITERATURE REVIEW

Recommendation systems have been widely employed in e-commerce to assist identifying relevant products from extensive product catalogs. Earlier work of Schafer, Konstan, and Riedl has discussed the application of recommendation techniques in the context of online shopping, as well as the utilization of such data to support the selection of products [6]. With the emergence of the e-commerce platforms, much

of the current literature has been concerned with session-based recommendation, where interactions within the current shopping session can be leveraged to represent the current interests of the customer.

The context of session-based recommendation is particularly useful when information about the customer's global profile is limited. Bauer and Jannach have investigated hybrid session-aware recommendation, based on session interactions plus additional item- and contextual features [14]. Salampasis *et al.* have studied different approaches to session-based recommendation in the context of e-commerce, including co-occurrence-based as well as more sophisticated recommendation techniques [7]. Kersbergen *et al.* have investigated session-based recommendation at scale, and emphasized the importance of low-latency recommendation generation in the context of e-commerce [8]. These works thus show that interactions within the current shopping session can be beneficial in capturing the customer's short-term preferences.

The task of purchase-intention prediction is closely related to the idea of session-based recommendation, yet its purpose is different. Whereas session-based recommendation approaches explicitly choose the next product to recommend, the purpose of purchase-intention prediction is to estimate the likelihood that the current session will conclude with a purchase intent. Esmeli *et al.* have investigated early purchase-intention prediction using online retail session data, and discussed the utilization of behavioral information before the end of the session [3]. Gomes *et al.* have presented customer representations generated from online shopping sessions for purchase-intention prediction purposes [2]. Tokuc, and Dag have compared different clickstream representations and machine learning models for the task of purchase prediction [1]. Their approach has considered aggregate session information, recent actions, and hybrid representations; these are relevant to the feature representation adopted in the proposed system.

Tree-based machine learning methods are widely applied in the context of structured prediction tasks. LightGBM, as introduced by Ke *et al.*, is a gradient boosting decision tree framework designed for efficient model training and prediction [9]. In the implemented system, LightGBM is combined with XGBoost and Random Forest, as opposed to being applied as a single predictor. The probability outputs from these base learners are provided to a Logistic Regression meta-model, as part of the stacking procedure. In this way, the final predictor can leverage the information obtained from multiple base learners.

An area of research closely related to purchase-intention prediction is that of anomaly detection. Isolation Forest, proposed by Liu, Ting, and Zhou, is able to identify unusual observations from the given set by isolating them from the remainder of the observations [10]. Much of the literature on e-commerce has investigated both behavioral and graph-based approaches to detect unusual and potentially fraudulent behavior. For example, eFraudCom has applied competitive graph neural networks to detect fraud in e-commerce, whereas behavioral graph-based approaches have exploited interaction

relationships to detect such behavior [11], [12]. On the other hand, the objective of Model 2 in the present work is different from these fraud-detection systems; it uses Isolation Forest to detect unusual brand-level behavioral patterns, and treats the resulting output as the behavioral suspiciousness/trust signal. Trust-related information can be considered to be relevant when designing recommendation systems. Cabrera-Sánchez *et al.* have investigated factors affecting the use of online recommendation systems in the context of e-commerce [13]. Some research on the topic of session-aware recommendation has also assessed the influence of popularity and feedback properties on offline recommendation evaluation [14]. These works thus provide motivation for considering information beyond product relevance when recommending products.

Based on the reviewed literature, the phenomenon of session behavior has been extensively analyzed both for recommendation and purchase-intention prediction. However, the context of anomaly detection has been mainly pursued as an endeavor to detect unusual e-commerce behavior. The implemented system thus connects these topics by utilizing the current-session purchase probability along with the brand-level behavioral trust when re-ranking recommendation candidates. Thus, the anomaly-related output is not perceived as an exclusive detection signal, but rather as one of the factors influencing the order of recommendation candidates.

3. METHODOLOGY

The proposed system is developed as a combination of purchase prediction, brand-level behavioral analysis, and recommendation ranking. The methodology is divided into three major parts. The first part estimates the probability that a user will make a purchase based on the actions performed during a session. The second part studies the behavior associated with different brands and generates suspiciousness and trust values. The third part uses these outputs along with product relevance to generate the final recommendation list.

The processing begins with the raw e-commerce clickstream data. The data is first cleaned and arranged into user sessions. Features are then extracted from the session activity and given to the purchase prediction model. At the same time, brand-level behavioral features are generated for the second model. Product candidates are obtained using different retrieval methods and are combined with the outputs from both models. The final candidates are assigned a trust-aware score and sorted to produce the recommendations.

A. Dataset and Preprocessing

The system uses the 2019-Oct.csv e-commerce clickstream dataset. Each record contains information such as event time, event type, product ID, category information, brand, price, user ID, and user session ID. These fields are used to understand the sequence of actions performed by users while browsing the platform.

Before model training, the dataset is cleaned to remove duplicate records. Missing brand values are represented as unknown, while the category-code field is not used because

Table 3.1
DATASET AND EVALUATION STATISTICS

Item	Validated Value
Raw clickstream file	2019-Oct.csv
Product catalog	30,223 products
Brand catalog / Model 2 coverage	765 brands
Held-out Model 1 sessions	1,069
Recommendation evaluation sessions	1,000
Recommendation candidates	29,571
Average candidates/session	29.57
Unique products in integrated candidates	4,960
Unique brands in integrated candidates	733

of the missing values present in that field. The event time is converted into datetime format. After this conversion, the records are sorted according to the user session and event time. This ordering is required because the sequence in which the user performs actions is considered during feature generation.

The validated product catalog contains 30,223 products from 765 brands. Model 2 produces trust information for these 765 scored brands. For recommendation testing, 1,000 sessions are used and 29,571 candidate rows are generated. This corresponds to an average of 29.57 candidate products for each evaluation session.

B. Sessionization and Purchase Label

The records are grouped using the user_session identifier. Within every session, the events are arranged according to their timestamp. This maintains the actual order of user activity and allows the model to distinguish between different sequences of actions.

Sessions with very few interactions are excluded since they do not contain enough information for reliable behavioral analysis. A session is labelled as a purchase session when at least one purchase event is present. Sessions without a purchase event are assigned the non-purchase label.

After the target label is generated, the purchase event is removed from the model input. This step is important because keeping the purchase event would directly reveal the outcome that the model is expected to predict.

C. Session Feature Engineering

The system does not treat every click as a separate training sample. Instead, the activities performed during a complete session are summarized into a set of features. This provides the model with information about the overall behavior of the user during that session.

The session features include the total number of events, number of product views, number of cart actions, number of unique products, number of unique categories, and number of unique brands. Price information is represented using statistics such as the average, minimum, and maximum product price observed in the session.

The time-related features include session duration, starting hour, day of the week, and a weekend indicator. The recent activity of the user is also included because the latest actions can provide useful information about purchase intent. The

five most recent non-purchase actions are stored as sequential features. A product view is represented by 1 and a cart action by 2. When fewer than five actions are available, the unused positions are filled with 0.

This combination of aggregate and recent-action features allows the model to consider both the general behavior of the session and the user's latest interactions.

D. User Intent Prediction Using LightGBM and Stacking

Four classifiers are implemented for predicting purchase intent: Logistic Regression, Random Forest, XGBoost, and LightGBM. LightGBM is configured with 200 estimators, a learning rate of 0.05, 31 leaves, and balanced class weighting. XGBoost uses 200 estimators, a learning rate of 0.05, and a maximum depth of 5. Random Forest is configured with 300 trees, a maximum depth of 12, and balanced class weighting.

The final purchase prediction is obtained using a stacking approach. LightGBM, XGBoost, and Random Forest are used as the base models. Their predictions are then given to Logistic Regression, which acts as the meta-model. Five-fold stratified cross-validation is used during training to generate out-of-fold probability values for the base models.

For a new session, each fold-specific base model produces a prediction. These predictions are averaged and passed to the Logistic Regression meta-model. The final result is a continuous purchase probability rather than only a binary purchase or non-purchase output.

The purchase probability for a session s is represented as $P(\text{purchase} | s)$ and is later used during recommendation ranking.

$$P(\text{purchase} | s) = \text{Stacking} \left(\begin{array}{c} \text{LightGBM}(s), \\ \text{XGBoost}(s), \text{RandomForest}(s) \end{array} \right) \quad (1)$$

E. Threshold Optimization

The probability generated by the stacking model is continuous, so a threshold is required only when the purchase prediction is evaluated as a binary classification problem. The recommendation stage continues use the original probability value.

A threshold of 0.5 was considered initially. However, different threshold values were tested because the dataset contains an imbalance between purchase and non-purchase sessions. Among the tested values, a threshold of 0.90 produced the highest F1-score.

At the selected threshold of 0.90, the purchase prediction model achieved an accuracy of 0.9371, precision of 0.6364, recall of 0.3500, and F1-score of 0.4516. ROC-AUC and PR-AUC are calculated directly from the probability outputs and therefore are not affected by the selected classification threshold.

F. Brand Behavioral Feature Engineering

The second model works with brand-level information rather than individual user sessions. The clickstream activity is

Table 3.2
BEHAVIORAL SUSPICIOUSNESS AND TRUST CATEGORIES

Risk Level	Brands	Percentage	Trust Range
Low	710	92.81%	0.7000–0.9995
Medium	49	6.41%	0.4000–0.6999
High	6	0.78%	0.1500–0.3999
Total	765	100%	0.1500–0.9995

aggregated for each brand to describe its observed behavioral pattern. Seven behavioral features are used for this purpose.

Before applying the anomaly detection model, the generated features are standardized using a fitted StandardScaler. Standardization puts the different feature values on a comparable scale and provides the processed input required by the anomaly detection stage.

The available validation information confirms that seven features are used, but it does not provide their individual feature names. Therefore, no additional feature names are specified here beyond the information available from the implementation.

G. Behavioral Suspiciousness Estimation Using Isolation Forest

Isolation Forest is used in Model 2 to identify brands that show unusual combinations of behavioral features. The implemented Isolation Forest contains 100 estimators and uses a contamination value of 0.08.

The output of this model is used to obtain suspiciousness and trust information for the recommendation stage. The purpose of this step is to identify unusual behavioral patterns rather than to directly declare a brand as fraudulent. Since the clickstream dataset does not contain verified fraud labels for all brands, the resulting suspiciousness value is treated as an indicator of unusual behavior.

The validated Model 2 output contains 765 scored brands. The suspiciousness values range from 0.0005 to 8500, while the trust values range from 0.1500 to 0.9995. Based on the stored trust and risk information, the brands are divided into Low, Medium, and High risk categories.

The exact transformation used in the implementation to convert the raw Isolation Forest output into the stored suspiciousness and trust values is not specified in the available validation material. Therefore, no additional transformation equation is introduced.

H. Candidate Product Retrieval

Once the purchase probability and brand-level information are available, the recommendation module creates a set of possible products for each session. Three retrieval methods are used to form this candidate set.

The first method retrieves products that have been co-viewed with products present in the current session. The second method retrieves products from related categories. The third method uses globally popular products so that the candidate set also contains products that may be relevant even when limited session information is available.

Each retrieval method produces a relevance value for the products it returns. If a product appears through more than one retrieval method, duplicate entries are removed using the session and product identifiers. The highest available relevance value is retained. The final source relevance value is represented by $R(s, p)$ and is restricted to the range $[0.1, 1.0]$.

I. Trust Integration and Fallback Handling

The outputs from the two models are combined with the candidate product data. Model 1 is connected to the candidate data using the `user_session` identifier. Model 2 is connected using the brand associated with each product.

The integrated data contains the purchase probability, trust score, and suspiciousness values in the range $[0, 1]$. During the validated evaluation, all 29,565 evaluated candidates received an actual purchase probability from Model 1.

Model 2 supplied real trust values for 26,309 candidates, which represents 88.99% of the evaluated candidates. For the remaining candidates, fallback values are used when sufficient brand-level behavioral information is not available.

Among the evaluated candidates, 2,460 candidates, or 8.32%, had unknown brand information. These candidates were assigned a trust value of 0.50. A further 796 candidates, or 2.69%, belonged to brands having fewer than five lifetime events and were assigned a trust value of 0.60.

These fallback values represent limited behavioral evidence. They are not used as evidence that the corresponding brands are fraudulent.

J. Trust-Aware Recommendation and Re-ranking

The final ranking combines product relevance with the predicted purchase probability and the trust information obtained from Model 2. A risk penalty is also included so that the behavioral information can affect the position of a product in the recommendation list.

The final score for a product p in session s is calculated as follows:

$$S(s, p) = R(s, p) [1 + \alpha P(\text{purchase} \mid s)] \times \text{Trust}(b)^\beta \Psi(b). \quad (2)$$

Here, $R(s, p)$ is the relevance value of product p for session s . The term $P(\text{purchase} \mid s)$ is the purchase probability obtained from Model 1. $\text{Trust}(b)$ represents the trust value of the brand b , and $\Psi(b)$ represents the penalty associated with its risk category. The implementation uses $\alpha = 1.0$ and $\beta = 1.0$.

The risk penalty is set to 1.00 for Low-risk brands, 0.75 for Medium-risk brands, and 0.20 for High-risk brands. Therefore, a product with high relevance and high purchase probability can receive a higher ranking, while lower brand trust or a higher risk category can reduce its final score.

The system uses a soft penalty rather than removing products completely. Products belonging to the High-risk category receive a stronger reduction in their score, whereas Medium-risk products receive a smaller reduction. This allows the recommendation process to retain candidate products while still considering the behavioral information generated by Model 2.

K. Overall Algorithm

The complete workflow of the proposed system can be summarized in the following steps:

- 1) Load the e-commerce clickstream dataset and perform the required preprocessing.
- 2) Group the events using the `user_session` identifier and arrange the events in chronological order.
- 3) Generate session-level features, including aggregate activity, price-related information, time information, and recent user actions.
- 4) Use the trained stacking model to obtain the purchase probability for each session.
- 5) Aggregate the available behavioral information for each brand and generate the corresponding suspiciousness and trust values using Isolation Forest.
- 6) Retrieve possible recommendation candidates using co-view, co-category, and global popularity information.
- 7) Join the candidate products with the purchase probability from Model 1 and the brand-level trust information from Model 2.
- 8) Apply the defined fallback values when reliable brand-level behavioral information is not available.
- 9) Calculate the trust-aware recommendation score for every candidate product.
- 10) Sort the candidate products according to their calculated scores.
- 11) Select the Top- K products from the ranked candidate list.
- 12) Evaluate the recommendations using relevance, trust, exposure, sensitivity, and ranking-based measures.

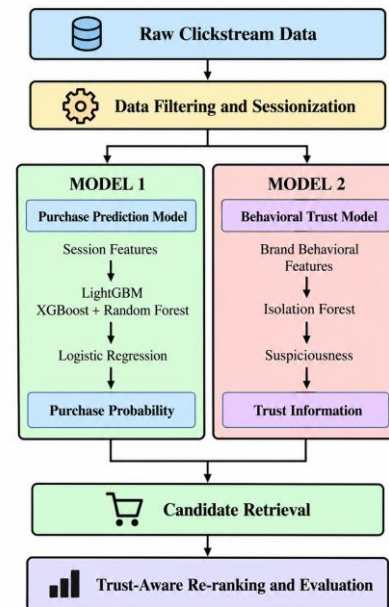


Figure 3.1. Overall System Architecture

4. RESULTS

The proposed system was evaluated to benchmark the performance of the purchase prediction model, the behavioral suspiciousness and trust model, and the final trust-aware recommendation pipeline. The evaluation also verifies that the outputs of both models are indeed being incorporated into the recommendation ranking process.

A. Dataset and Evaluation Statistics

The recommendation system evaluated on 1,000 sessions, which produced 29,571 candidate rows during the candidate generation stage. Both the purchase prediction and behavioral suspiciousness models were applied and their scores used to derive the final recommendation rankings.

Table 4.1
DATASET AND EVALUATION STATISTICS

Metric	Value
Evaluation Sessions	1,000
Candidate Rows	29,571

B. Purchase Prediction Results

Four implemented models were evaluated for purchase prediction: LightGBM, XGBoost, Random Forest, and the Stacking Ensemble. Their performance was measured using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and log loss. The classification performance results are presented in Table 4.2, while the ROC-AUC, PR-AUC, and log loss results are presented in Table 4.3.

Table 4.2
PURCHASE PREDICTION MODEL RESULTS: ACCURACY AND CLASSIFICATION METRICS

Model	Accuracy	Precision	Recall	F1
LightGBM	0.8483	0.2470	0.5125	0.3333
XGBoost	0.8316	0.2316	0.5500	0.3259
Random Forest	0.8945	0.3300	0.4125	0.3667
Stacking Ensemble	0.7946	0.1966	0.5750	0.2930

Table 4.3
PURCHASE PREDICTION MODEL RESULTS: ROC-AUC, PR-AUC AND LOG LOSS

Model	ROC-AUC	PR-AUC	Log Loss
LightGBM	0.7812	0.3695	0.3580
XGBoost	0.7791	0.3494	0.4108
Random Forest	0.7814	0.3664	0.3342
Stacking Ensemble	0.7845	0.3861	0.5244

The results show that the models differ across the evaluated performance metrics. Random Forest recorded the highest accuracy, precision, and F1-score among the individual models, while XGBoost recorded the highest recall. The Stacking

Ensemble recorded the highest ROC-AUC and PR-AUC in the comparison, although its accuracy and precision were lower than those of Random Forest.

For the final binary classification, an optimized decision threshold is considered, rather than the default threshold. The selected threshold along with its precision, recall, and F1-score is presented in Table 4.4.

Table 4.4
OPTIMIZED PURCHASE CLASSIFICATION THRESHOLD

Threshold	Precision	Recall	F1-score
0.90	0.6364	0.3500	0.4516

At the selected threshold of 0.90 the optimized classifier achieved an accuracy of 0.9371, precision of 0.6364, recall of 0.3500, and F1-score of 0.4516. Increasing the threshold makes the classifier more selective, and thus improves precision but decreases recall.

For the recommendation generation process, the continuous stacking probability is maintained, rather than converting it to a binary decision, so that the estimated purchase intent contributes directly to the ranking score.

C. Behavioral Suspiciousness and Trust Results

The second model operates at the level of brands, providing behavioral suspiciousness information, which is further converted into a trust value that informs the recommendation component. The resulting brands are divided into low-, medium- and high-risk behavioral categories.

Six brands in the high-risk category were identified by evaluation, as shown in Table 4.5.

Table 4.5
BEHAVIORAL RISK DISTRIBUTION

Risk Category	Number of Brands
High Risk	6

These categories represent the model's generated behavioral classification. Thus, a high-risk category reflects unusual behavioral characteristics identified by the implemented model, and should not be interpreted as a legal conclusion or proof of fraudulent activity.

D. Evidence That Both Models Affect the Recommendation

An integration check was performed to verify that the actual outputs of both models were used during the recommendation generation process. This served to also determine whether predefined fallback values were having a major influence on the final ranking.

The integration results are presented in Table 4.6.

Table 4.6
MODEL INTEGRATION AND RANKING INFLUENCE

Integration Measure	Result
Model 1 real probability usage	29,565/29,565 (100.00%)
Model 1 fallback intent usage	0.00%
Model 2 real trust usage	26,309/29,565 (88.99%)
Model 2 fallback trust usage	3,256/29,565 (11.01%)
Unknown-brand fallback	2,460 candidates (8.32%)
Low-confidence-brand fallback	796 candidates (2.69%)
Candidate rank positions displaced by trust	2,972
High-trust items promoted into top 3	628 instances
Medium/high-risk items demoted	1,623 instances

All 29,565 evaluated candidates received an actual purchase probability from Model 1, and therefore no fallback intent value was required. Model 2 supplied actual trust values for 26,309 candidates or 88.99% of the evaluated candidates.

For the remaining candidates, fallback values were required since certain brands were unknown or did not have sufficient evidence. The trust component changed 2,972 candidate ranking positions. Additionally, 628 high-trust items were promoted into the top three positions, while 1,623 medium or high-risk items were demoted. These results show that the trust information has a direct influence on final ranking.

E. Recommendation Baseline Comparison

The trust-aware ranking method was compared against popularity-based ranking, purchase-only ranking, and hard-filter ranking. The comparison considered Top-5 and Top-10 recommendations using HitRate, Precision, Recall, NDCG, Average Trust, and High-Risk Exposure. The results for popularity-based and purchase-only ranking are presented in Table 4.7, while the results for hard-filter and trust-aware ranking are presented in Table 4.8.

Table 4.7
RECOMMENDATION BASELINE COMPARISON: POPULARITY AND PURCHASE-ONLY

Metric	Popularity	Purchase-only
HitRate@5	0.1400	0.7630
Precision@5	0.0280	0.1554
Recall@5	0.1400	0.7630
NDCG@5	0.0896	0.4826
Average Trust@5	0.5629	0.6158
High-Risk Exposure@5	1.16%	1.36%
HitRate@10	0.2050	0.9210
Precision@10	0.0207	0.0937
Recall@10	0.2050	0.9210
NDCG@10	0.1107	0.5349
Average Trust@10	0.5752	0.5999
High-Risk Exposure@10	0.98%	1.21%

Table 4.8
RECOMMENDATION BASELINE COMPARISON: HARD-FILTER AND TRUST-AWARE

Metric	Hard-filter	Trust-aware
HitRate@5	0.7520	0.8220
Precision@5	0.1532	0.1672
Recall@5	0.7520	0.8220
NDCG@5	0.4759	0.5685
Average Trust@5	0.6223	0.6677
High-Risk Exposure@5	0.00%	0.00%
HitRate@10	0.9060	0.9410
Precision@10	0.0922	0.0956
Recall@10	0.9060	0.9410
NDCG@10	0.5269	0.6078
Average Trust@10	0.6062	0.6654
High-Risk Exposure@10	0.00%	0.00%

For both $K = 5$ and $K = 10$, the trust-aware approach recorded the highest measured values for HitRate, Precision, Recall, and NDCG among the evaluated ranking strategies. At $K = 10$, NDCG increased from 0.5349 with purchase-only ranking to 0.6078 with trust-aware ranking.

Additionally, Average Trust@10 increased from 0.5999 to 0.6654. The hard-filter approach recorded zero high-risk exposure, yet its HitRate@10 and NDCG@10 values were lower than those obtained using the trust-aware method.

The comparison shows the difference between completely removing candidates and reducing their ranking priority. The implemented trust-aware approach applies a soft penalty, allowing relevance and purchase intent to remain part of the ranking, while behavioral trust influences the final position.

F. Fallback-Only Ablation

An additional ablation evaluated a fallback-only configuration against the configuration using the actual machine learning outputs. The results are presented in Table 4.9.

Table 4.9
FALLBACK-ONLY ABLATION

Metric	Fallback-only	Real ML Models	Change
HitRate@10	0.9210	0.9410	+0.0200
NDCG@10	0.5349	0.6078	+0.0729
Average Trust@10	0.6000	0.6999	+0.0999
High-Risk Exposure@10	1.21%	0.00%	-1.21 pp

Using the actual machine learning outputs increased HitRate@10 by 0.0200 and NDCG@10 by 0.0729, while Average Trust@10 increased by 0.0999, and High-Risk Exposure@10 decreased from 1.21% to 0.00%.

The ablation therefore provides additional evidence that the recommendation results are influenced by the actual model predictions rather than being generated predominantly from fixed fallback values.

G. High-Risk Ranking Case Study

A controlled example involving three products was used to illustrate the impact of the trust signal on the final ranking.

All three candidates were assigned the same relevance value of 1.0 and purchase probability of 0.80, while their trust values differed according to their behavioral risk categories.

The “globber” brand had a trust value of 0.198 and was categorized as High Risk, “samsung” was assigned a trust value of 0.582 and categorized as Medium Risk, and finally “apple” had a trust value of 0.999 and was categorized as Low Risk.

Without the trust component, all three candidates received the same purchase-only score of 1.8000. Once the trust-aware ranking calculation was applied, their scores became 0.0713, 0.7857, and 1.7982 for globber, samsung, and apple respectively. The resulting ranking scores are illustrated in Fig. 4.1.

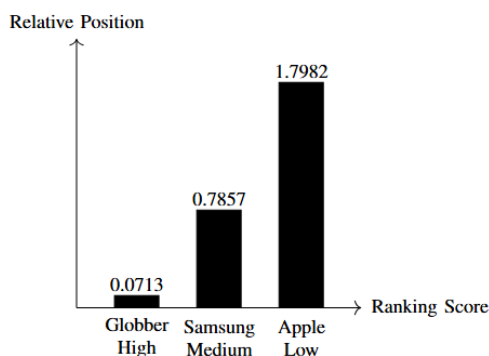


Figure 4.1. Effect of Trust on a Controlled Ranking Example

The example shows that the trust component has the ability to change the relative ordering of candidates, even with identical relevance and purchase probability.

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5. FUTURE DISCUSSION / FUTURE WORK

The current implementation offers an end-to-end offline trust-aware recommendation pipeline, but there are certain ways in which it could be improved before deployment in a real e-commerce environment.

First, behavioral modelling could be enhanced by using longer and more detailed interaction histories. Instead of only using aggregated brand behavior, future versions could model how behavior changes over time and how patterns differ across product categories. This could reduce false suspiciousness for brands that naturally have unusual but legitimate activity.

Second, additional data sources could be included. Product reviews, return behavior, seller history, price changes, delivery information and verified transaction outcomes could provide stronger evidence than clickstream behavior alone. These signals could be added with care so that the trust model is not biased towards brands with higher traffic.

Third, the recommendation model could be extended beyond the current multi-channel candidate retrieval and multiplicative re-ranking formula. Learning-to-rank methods, neural session models, graph-based recommendation and hybrid rec-

ommenders could be evaluated while retaining the trust signal as an additional ranking objective.

Fourth, the system could be moved towards real-time recommendation. The current validation processes batches of session data and saved model outputs. A production version could update session features after every interaction and retrieve new candidates with low latency. The brand trust model could also be updated periodically as new behavioral evidence becomes available.

Fifth, trust estimation should be calibrated and validated with stronger ground truth. Human review, verified seller information and confirmed platform outcomes could help determine whether high suspiciousness actually corresponds to meaningful risk. Until such evidence is available, the system should continue to describe the output as behavioral suspiciousness or trust estimation rather than fraud detection.

Finally, explainability should be added to the recommendation interface. A user or platform administrator should be able to see why an item was ranked highly, for example because it matched the current session and had a high purchase probability, while another item was demoted because its associated brand had a lower behavioral trust score. This would enable the system to be more easily inspected and improved.

6. CONCLUSION

The proposed system has been created and validated as an e-commerce recommendation system that leverages clickstream data to combine purchase intent and behavioral trust. The purchase prediction component synthesizes the session behavior into aggregate and recent-action features and applies LightGBM, XGBoost and Random Forest inside a stacking ensemble. The second component uses Isolation Forest on brand-level behavioral features to provide suspiciousness and trust information which are later integrated to trusted candidate retrieval and trust-aware re-ranking.

The experimental results demonstrate that the proposed integration is indeed a practice rather than a theory. All the evaluated recommendation candidates received real Model 1 purchase probabilities, 88.99% got real Model 2 trust values and trust resulted in thousands of candidate rank changes. The proposed ranking has achieved NDCG@10 of 0.6078 compared with 0.5349 for purchase-only ranking and had resulted in 0.00% High-Risk Exposure@10 in the reported baseline comparison. The system therefore can be considered as a practical way to include behavioral trust information in e-commerce recommendation while keeping an adequate distinction between unusual behavior and proven fraud.

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CONFERENCE APPLICATION RECORD

Paper Title	Trust Aware E-Commerce Recommendation Using Brand Risk Analysis
Authors	Ms. M. Shafiya Banu — Mentor; Shruthi S — Student; Shahnaz Fathima N — Student
Affiliation	Department of Computer Science and Engineering, Rajalakshmi Engineering College
Conference Name	International Conference on AI and Machine Learning for Big Data Analytics (ICAML-26)
Conference Type	International Conference
Application / Submission Status	Abstract successfully received and currently under review
Conference Date	21st September–22nd September 2026
Submission Confirmation	1 September 2026, 10:19 AM
Location	Leipzig, Germany
Relevant Topic / Domain	AI and Machine Learning for Big Data Analytics
Abstract / Article ID	WRF-9-1-0090890
Conference Contact	abstracts@wrfconference.com; Phone: +919940951754

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Figure: Conference Submission/Application Evidence

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Authors	Ms. M. Shafiya Banu — Mentor; Shruthi S — Student; Shahnaz Fathima N — Student
Affiliation	Department of Computer Science and Engineering, Rajalakshmi Engineering College
Conference Name	International Conference on AI-driven Data Analytics and Knowledge Discovery (ICADK-26)
Conference Type	International Conference
Application / Submission Status	Research article accepted for presentation; conference registration requested
Conference Date	23/09/2026 to 24/09/2026
Notification Date	3 September 2026, 11:01 AM
Location	Paris, France
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